

12 September 2026

Exposure-scaled injection study

Conditional full-statistics peak catalogue for 2021

This catalogue asks what selected positive structures in the nominal 2021 1% and 10% samples could look like at full statistics *if their fitted excesses were persistent signals*. The two source samples motivate separate Monte Carlo options: $1\% \times 100$ and $10\% \times 10$.

Why local significance scales as the square root of exposure

Let f be the observed exposure fraction and $k = 1/f$. For unchanged selection, detector response and signal-to-background ratio, the signal and background means scale together. In the statistically dominated, small-signal limit,

$$s_{100\%} = k s_f, \quad b_{100\%} = k b_f, \quad (1)$$

$$Z_{100\%} \simeq \frac{k s_f}{\sqrt{k b_f}} = \sqrt{k} Z_f. \quad (2)$$

The requested reference significances are therefore

$$\boxed{Z_{1\% \rightarrow 100\%}^{\text{target}} = 10 Z_{1\%}^{\text{obs}}} \quad \boxed{Z_{10\% \rightarrow 100\%}^{\text{target}} = \sqrt{10} Z_{10\%}^{\text{obs}}}. \quad (3)$$

For independent Poisson bins with *known* backgrounds, the Asimov significance obeys the stronger identity

$$Z_{A, \text{known } b}^2(s, b) = 2 \sum_i \left[(s_i + b_i) \ln \left(1 + \frac{s_i}{b_i} \right) - s_i \right], \quad (4)$$

$$Z_{A, \text{known } b}(ks, kb) = \sqrt{k} Z_{A, \text{known } b}(s, b). \quad (5)$$

This follows directly by substitution and does not require $s_i \ll b_i$. The actual fits here profile a learned Gaussian-process (GP) background; their response need not satisfy this identity. The Asimov and likelihood conventions follow Cowan et al. [1].

Two amplitudes answer two different questions

For a positive source yield $\hat{a}_f$, source GP continuum $\hat{b}_{f,i}$, and bin-integrated Gaussian template $t_i(m_0)$, define

$$a_{\text{yield}} = k \max(\hat{a}_f, 0), \quad \lambda_i(a) = k \hat{b}_{f,i} + a t_i(m_0), \quad (6)$$

$$Z_A^{\text{GP}}(a_{\text{match}}; m_0) = \sqrt{k} Z_f^{\text{obs}}, \quad R_a = \frac{a_{\text{match}}}{a_{\text{yield}}}. \quad (7)$$

The yield-scaled amplitude is evaluated once on its deterministic Asimov spectrum. The 20 toys use a_{match} , solved numerically so that the fixed-mass GP-profile Asimov score reaches the requested target. The catalogue reports both $Z_A^{\text{GP}}(a_{\text{yield}})$ and R_a ; matching the target is a construction of the injection, not a test that the fit obeys ideal scaling.

What this conditional picture can establish

An upward fluctuation does not become a growing signal merely by collecting more data. Selecting observed peaks favors overestimates of any underlying yield. These injections illustrate a persistent-excess, upper-envelope idea; they do not define a statistical upper bound, a full-data prediction interval, or a discovery probability. No observed 100% 2021 spectrum enters this study.

Source choices, fits and the meaning of a maximum

The two source-conditioned options

Option	Historical nominal 1%	Native nominal 10%
Count factor	$k = 100$	$k = 10$
Source input	<code>final_1pct_invM.root</code>	Pinned 2021 10% spectrum
GP support	40–300 MeV	36–300 MeV
Kernel state	Own reviewed historical scan	Own reviewed released states
Target	$10Z_{1\%}^{\text{obs}}$	$\sqrt{10}Z_{10\%}^{\text{obs}}$

The 1% file is the input last used in the historical residual-truth studies, as requested. Its cuts, trigger-category membership, event overlap and exact effective exposure relative to the 10% sample remain unverified. Consequently, this is a comparison of two nominal exposure projections, not a measurement of exposure scaling for an identical event selection. **The source score Z_f is a fresh v5.6 evaluation of the observed histogram at archived kernel coordinates, not a verbatim archived significance.** Differences from older profile calculations are not exposure effects.

Selection and numerical model

Both observed sources are scanned from 50 to 250 MeV in 1 MeV steps. Positive regional maxima are selected in the predeclared intervals 60–70, 71–83, 86–99, 111–132, 148–177 and 198–223 MeV. Additional 1% entries come from the six largest interior local maxima, adding each distinct mass not already selected. Nearby alternatives, including 82 and 84 MeV, have correlated source structure and must not be combined. The documented support-sensitive 1% results at 50–52 MeV are excluded. Each entry records its actual selected mass; a regional endpoint is explicitly marked and need not be an interior peak. Selection precedes toy generation.

Counts use 0.625 MeV bins and the inherited fully scaled 2021 mass-resolution polynomial. The template t_i is the Gaussian probability integrated over bin i ; its amplitude is the full Gaussian yield. At every test mass, the GP is conditioned on bins outside $\pm 2.25\sigma_m$ using that source’s fixed reviewed kernel coordinates. Poisson noise, sideband conditioning, background prediction and covariance are recalculated for every toy. Hyperparameter uncertainty and optimization variability are omitted. The full-support Poisson generator includes signal tails in the sidebands.

For the extracted bins, b_i is the conditioned GP mean and LL^T its regularized count covariance. The profiled log likelihood is

$$\ell(a, \theta; m) = \sum_i [n_i \ln \lambda_i - \lambda_i] - \frac{1}{2} \theta^T \theta, \quad \lambda_i = b_i + (L\theta)_i + a t_i(m), \quad (8)$$

with positive fitted means. A negative unconstrained yield is retained when finding the best fit; the one-sided local statistic is

$$q_0(m) = \begin{cases} 2[\ell(\hat{a}, \hat{\theta}; m) - \ell(0, \hat{\theta}_0; m)], & \hat{a} \geq 0, \\ 0, & \hat{a} < 0, \end{cases} \quad Z_{\text{local}}(m) = \sqrt{q_0(m)}. \quad (9)$$

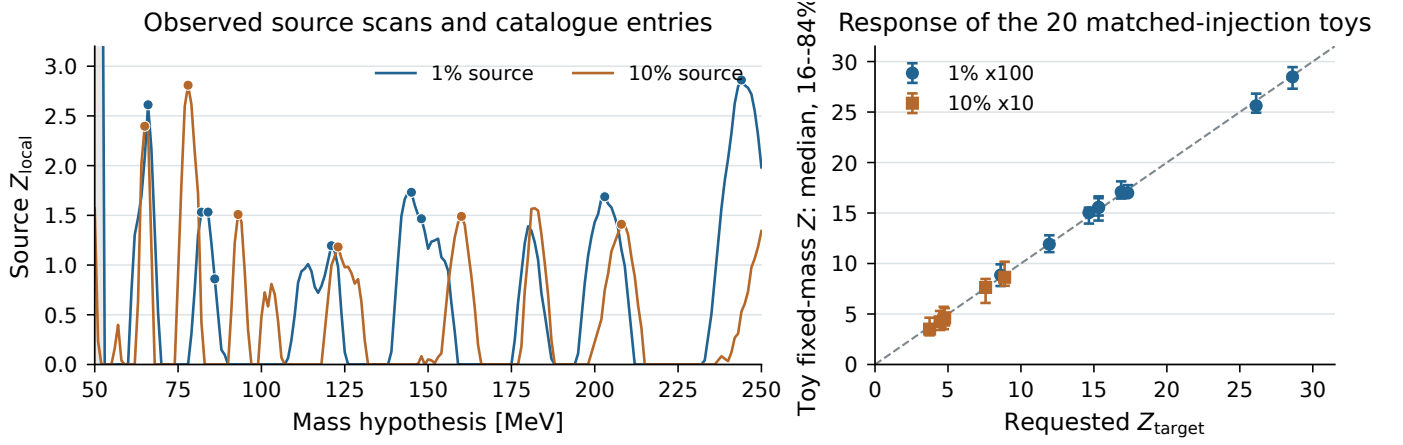
These are nominal, fixed-mass asymptotic scores. Each toy also has a nearby maximum on the integer grid within $\pm 2\sigma_m$ of the injection, bounded to 50–250 MeV. That maximum is tabulated separately and is not a global significance.

Twenty toys illustrate an upper realization, not an upper limit

For iid fixed-mass scores with CDF F , the maximum obeys $\Pr(M_{20} \leq z) = F(z)^{20}$. Under the additional illustrative approximation $Z \sim N(Z_A, 1)$, its median is $Z_A + \Phi^{-1}(2^{-1/20}) \simeq Z_A + 1.82$. This is not a validated law for these GP fits or the nearby-mass maxima. The empirical 16–84% interval and observed maximum below describe only 20 realizations per scenario; threshold counts cannot calibrate rare tails.

The catalogue contains 15 scenarios and 300 saved toys.

Catalogue overview



Left: observed source scores; dots mark selected entries and the gray strip marks the excluded 1% edge region. Right: fixed-mass toy medians with empirical 16–84% ranges. The diagonal is a visual reference.

Source	Region [MeV]	m_0 [MeV]	Z_f	Target	Z_A^{yield}	R_a	Toy median [16,84%] fixed mass	Max ₂₀ fixed mass
1%	65	66	2.61	26.12	23.43	1.12	25.62 [24.93, 26.82]	27.79
10%	65	65	2.40	7.58	6.76	1.12	7.63 [6.10, 8.47]	9.59
1%	75	82	1.53	15.32	13.99	1.10	15.54 [14.25, 16.47]	17.92
10%	75	78	2.81	8.88	8.39	1.06	8.64 [7.80, 10.18]	11.62
1%	92	86 [†]	0.86	8.61	7.99	1.08	8.86 [7.77, 9.92]	10.45
10%	92	93	1.51	4.77	4.45	1.07	4.63 [3.92, 5.58]	6.33
1%	120	121	1.19	11.94	11.13	1.07	11.91 [11.13, 12.79]	13.04
10%	120	123	1.18	3.74	3.64	1.03	3.51 [2.89, 4.63]	4.92
1%	160	148 [†]	1.47	14.66	13.59	1.08	15.03 [13.96, 15.53]	16.53
10%	160	160	1.49	4.71	4.54	1.04	4.39 [3.50, 5.71]	7.35
1%	210	203	1.69	16.87	15.50	1.09	17.09 [16.43, 18.13]	19.02
10%	210	208	1.41	4.46	4.16	1.07	4.23 [3.41, 5.30]	6.90
1%	+84	84	1.53	15.33	14.05	1.09	15.57 [14.75, 16.65]	17.05
1%	+145	145	1.73	17.32	16.07	1.08	16.98 [16.78, 17.76]	18.77
1%	+244	244	2.86	28.61	25.31	1.13	28.48 [27.31, 29.45]	30.30

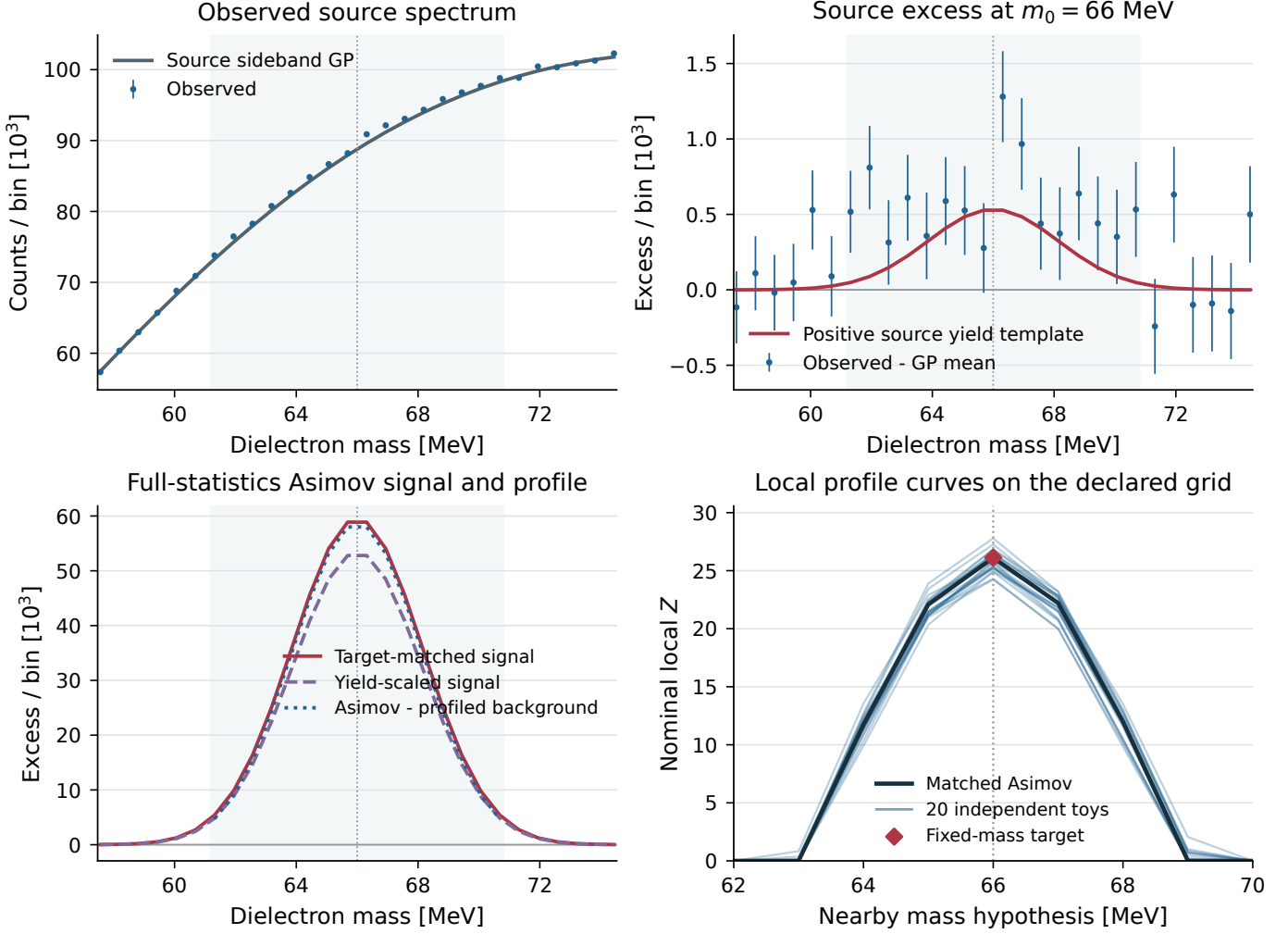
†: maximum at an endpoint of the source selection interval. “+” denotes an additional 1% peak. Z_A^{yield} is the deterministic response to $k\hat{a}_f$; all toy summaries use the target-matched amplitude. Max₂₀ is the largest of the 20 fixed-mass scores.

How to read the following pages

The matched-to-yield amplitude ratio spans 1.03–1.13. Values away from one quantify how much the injection must change to reach the requested GP-profile score; they should not be interpreted as a physical enhancement in the source yield. Each requested region is shown as adjacent 1% and 10% entries. The first page shows the observed spectrum, source excess, deterministic injection and all local profile curves. The second shows every saved toy, in seed order, with common mass and residual axes within the scenario. The residual panels subtract the known generator continuum to make morphology visible; their Poisson error bars do not include background-model uncertainty.

65 MeV region: $1\% \times 100$

Injection at $m_0 = 66$ MeV, $\sigma_m = 2.132$ MeV; source $Z_f = 2.612$, requested $Z_{\text{target}} = 26.119$.



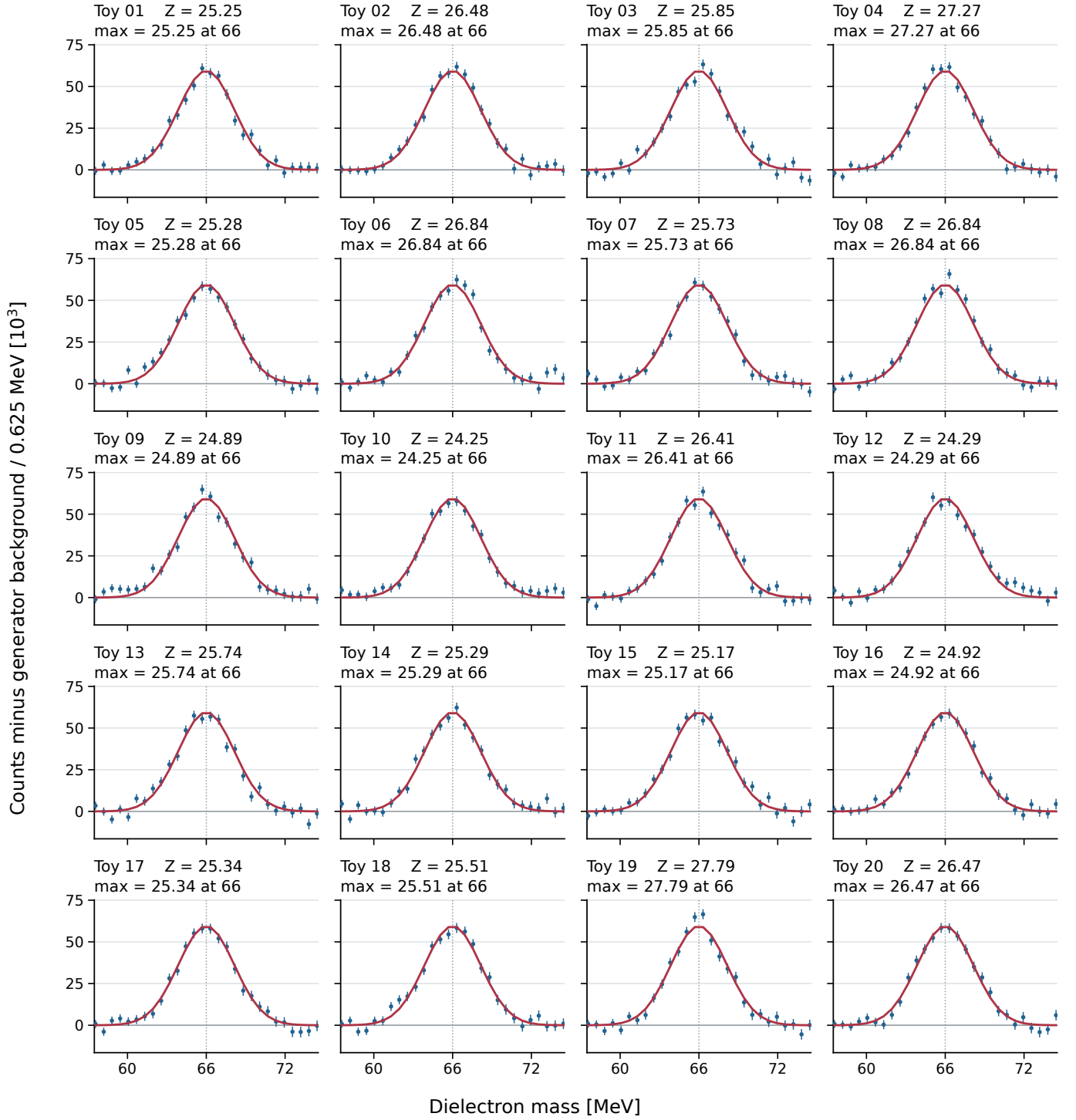
Shaded bands mark the $\pm 2.25\sigma_m$ extraction window. Observed error bars are $\sqrt{n}$; the dotted Asimov excess subtracts the profiled background in that window. The lower-right curves use fresh GP conditioning at every tested mass.

Yield-scaled amplitude a_{yield}	457,749	Matched amplitude a_{match}	510,570
Yield-scaled Asimov Z	23.433	Amplitude ratio R_a	1.115
Background-only Asimov Z	0.127	Matched Asimov $Z(m_0)$	26.119
Signal inside fit window	500,873	Background inside fit window	140,220,917
Fixed-mass median Z	25.62	Empirical 16–84% interval	[24.93, 26.82]
Largest fixed-mass Z	27.79	Largest nearby-maximum Z	27.79
Fixed-mass $Z \geq 3$	20/20	Fixed-mass $Z \geq 5$	20/20

The nearby scan covers 62–70 MeV in 1 MeV steps (9 hypotheses). Its median maximum is 25.62; 0/20 toy maxima lie at a grid endpoint. The matched Asimov maximum is 26.12 at 66 MeV.

65 MeV region: $1\% \times 100$: 20 realizations

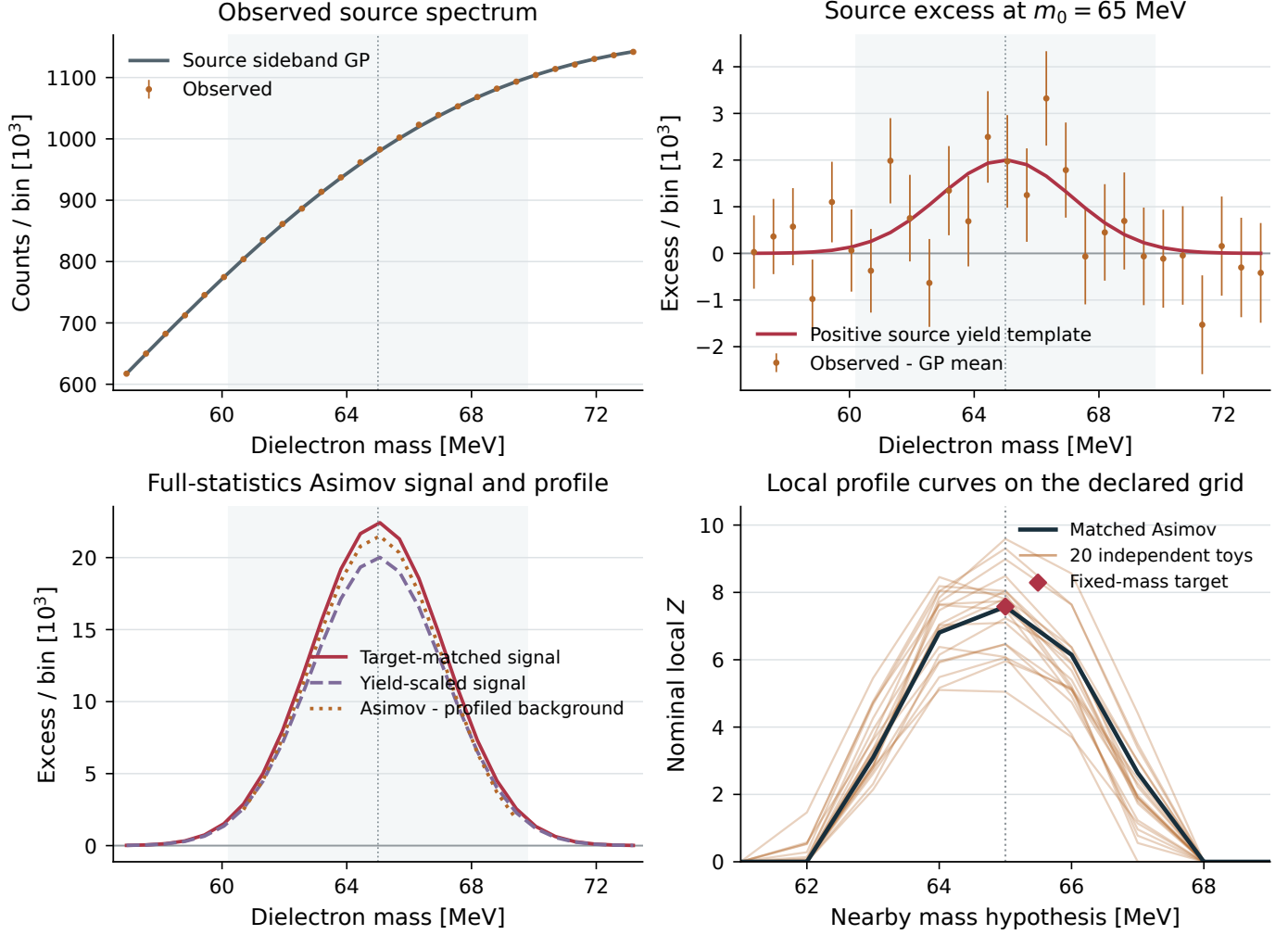
$1\% \times 100$; $m_0 = 66$ MeV; $Z_{\text{target}} = 26.12$. Red curves show the injected signal; points show counts minus generator background.



All 20 independent Poisson draws appear in saved seed order (display toy 01 is stored ID 0). Every panel uses identical axes within this scenario and $\sqrt{n}$ count errors. “ Z ” is evaluated at the injection mass; “max” scans the declared nearby hypotheses, with its mass in MeV. An asterisk marks a grid-endpoint maximum.

65 MeV region: $10\% \times 10$

Injection at $m_0 = 65$ MeV, $\sigma_m = 2.122$ MeV; source $Z_f = 2.396$, requested $Z_{\text{target}} = 7.577$.



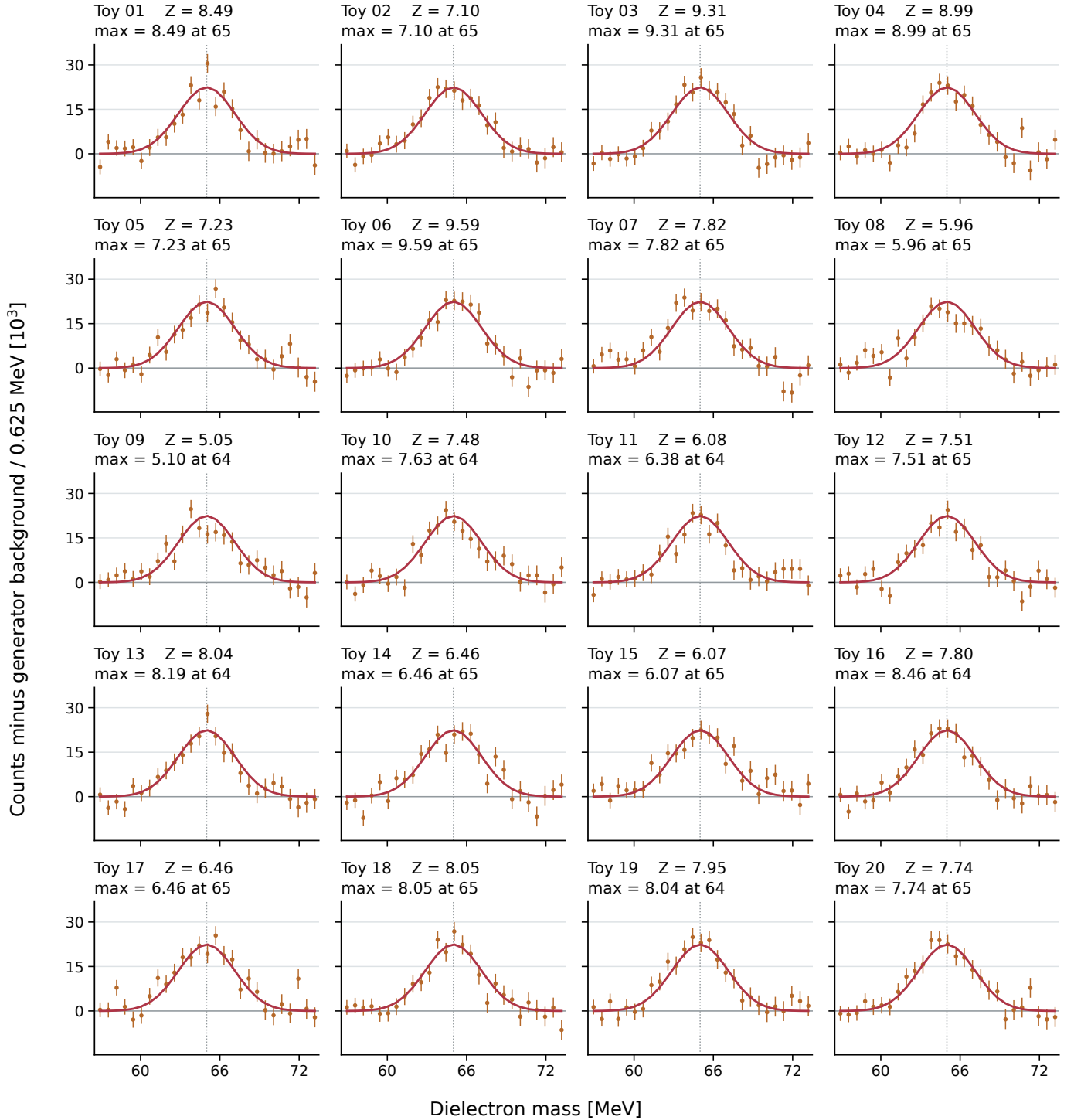
Shaded bands mark the $\pm 2.25\sigma_m$ extraction window. Observed error bars are $\sqrt{n}$; the dotted Asimov excess subtracts the profiled background in that window. The lower-right curves use fresh GP conditioning at every tested mass.

Yield-scaled amplitude a_{yield}	170,994	Matched amplitude a_{match}	191,565
Yield-scaled Asimov Z	6.758	Amplitude ratio R_a	1.120
Background-only Asimov Z	0.000	Matched Asimov $Z(m_0)$	7.577
Signal inside fit window	186,350	Background inside fit window	145,268,971
Fixed-mass median Z	7.63	Empirical 16–84% interval	[6.10, 8.47]
Largest fixed-mass Z	9.59	Largest nearby-maximum Z	9.59
Fixed-mass $Z \geq 3$	20/20	Fixed-mass $Z \geq 5$	20/20

The nearby scan covers 61–69 MeV in 1 MeV steps (9 hypotheses). Its median maximum is 7.68; 0/20 toy maxima lie at a grid endpoint. The matched Asimov maximum is 7.58 at 65 MeV.

65 MeV region: $10\% \times 10$: 20 realizations

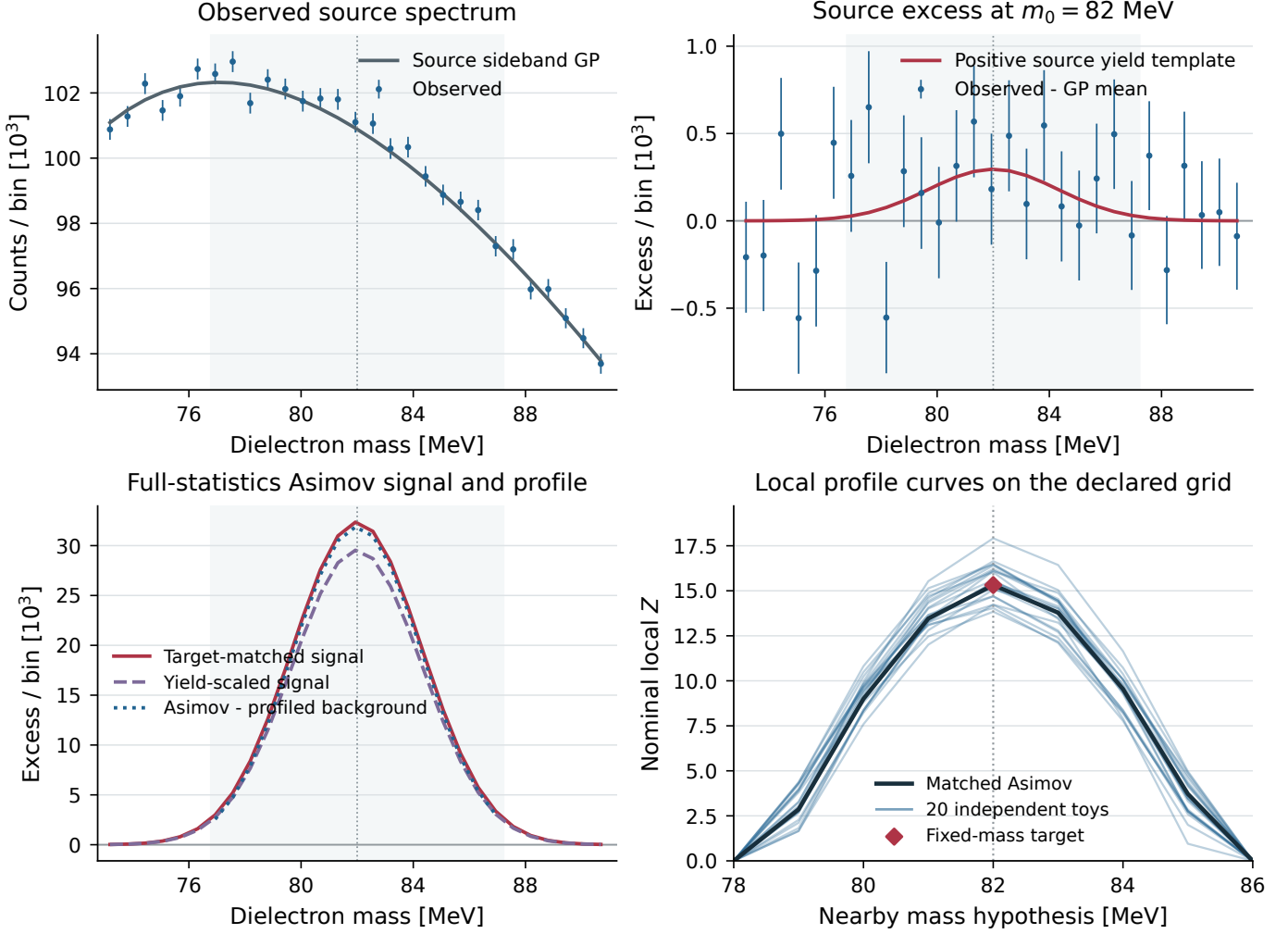
$10\% \times 10$; $m_0 = 65$ MeV; $Z_{\text{target}} = 7.58$. Red curves show the injected signal; points show counts minus generator background.



All 20 independent Poisson draws appear in saved seed order (display toy 01 is stored ID 0). Every panel uses identical axes within this scenario and $\sqrt{n}$ count errors. “Z” is evaluated at the injection mass; “max” scans the declared nearby hypotheses, with its mass in MeV. An asterisk marks a grid-endpoint maximum.

75 MeV region: $1\% \times 100$

Injection at $m_0 = 82$ MeV, $\sigma_m = 2.313$ MeV; source $Z_f = 1.532$, requested $Z_{\text{target}} = 15.318$.



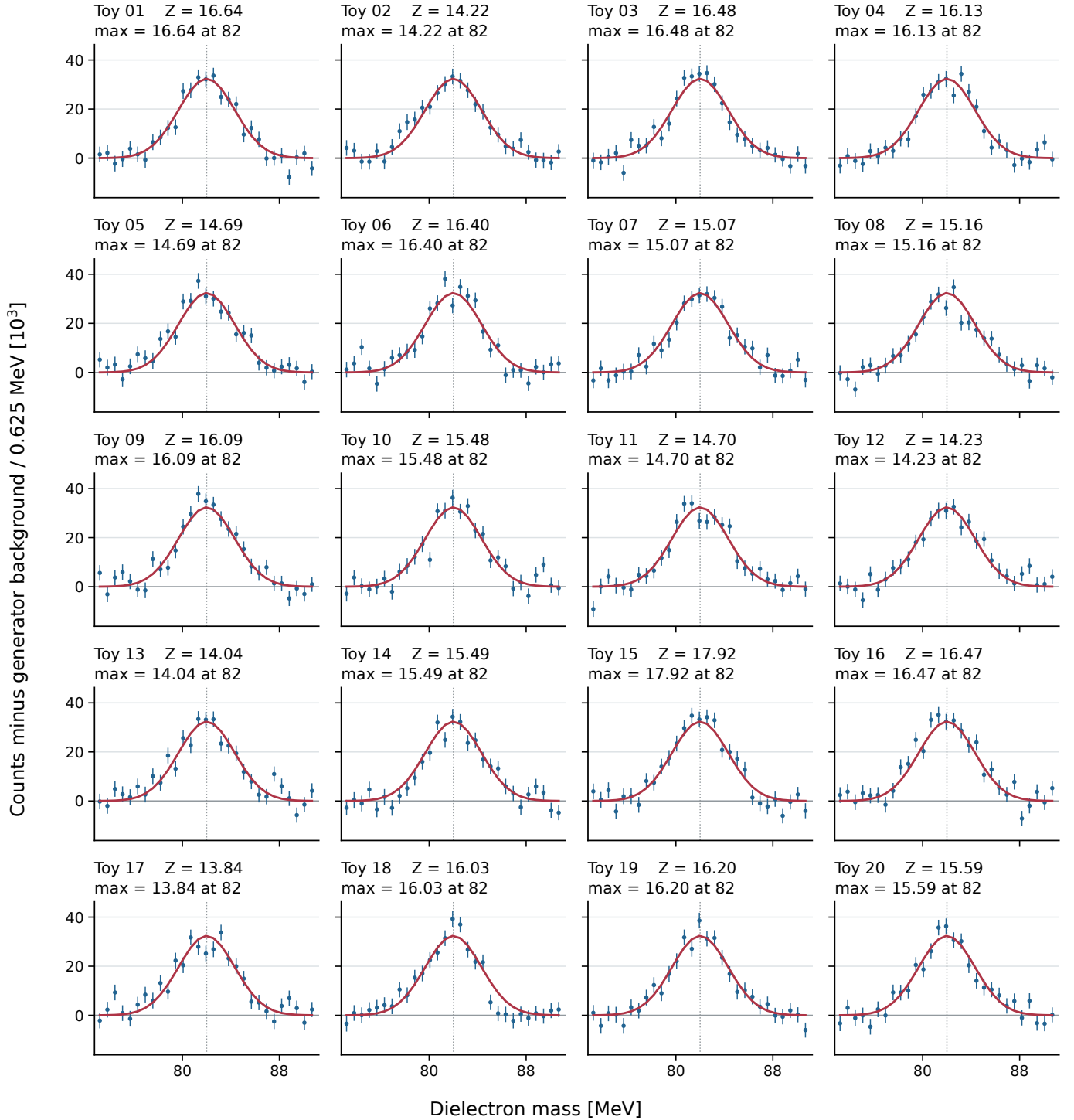
Shaded bands mark the $\pm 2.25\sigma_m$ extraction window. Observed error bars are $\sqrt{n}$; the dotted Asimov excess subtracts the profiled background in that window. The lower-right curves use fresh GP conditioning at every tested mass.

Yield-scaled amplitude a_{yield}	274,957	Matched amplitude a_{match}	301,154
Yield-scaled Asimov Z	13.993	Amplitude ratio R_a	1.095
Background-only Asimov Z	0.069	Matched Asimov $Z(m_0)$	15.318
Signal inside fit window	294,627	Background inside fit window	170,892,345
Fixed-mass median Z	15.54	Empirical 16–84% interval	[14.25, 16.47]
Largest fixed-mass Z	17.92	Largest nearby-maximum Z	17.92
Fixed-mass $Z \geq 3$	20/20	Fixed-mass $Z \geq 5$	20/20

The nearby scan covers 78–86 MeV in 1 MeV steps (9 hypotheses). Its median maximum is 15.54; 0/20 toy maxima lie at a grid endpoint. The matched Asimov maximum is 15.32 at 82 MeV.

75 MeV region: $1\% \times 100$: 20 realizations

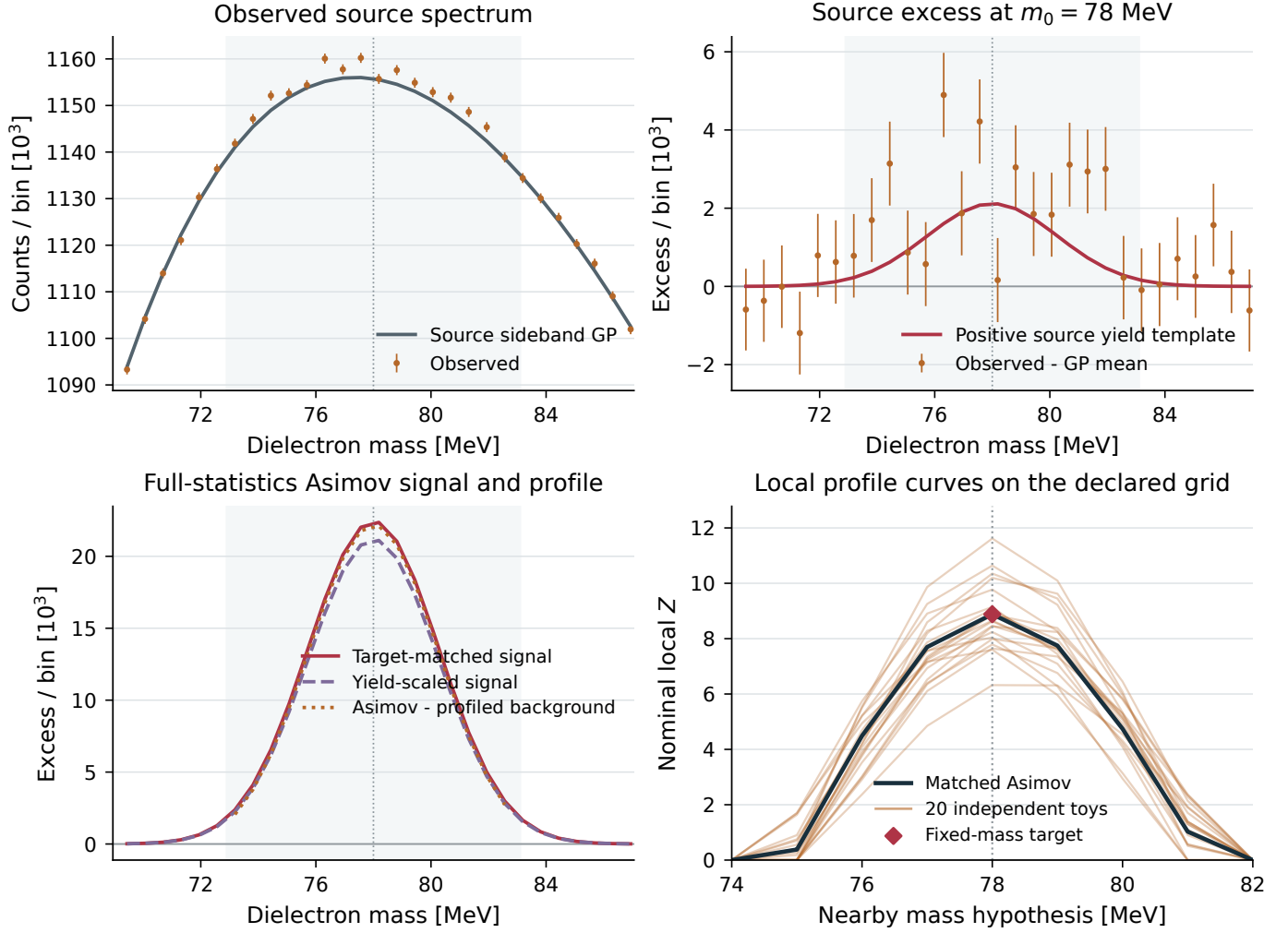
$1\% \times 100$; $m_0 = 82$ MeV; $Z_{\text{target}} = 15.32$. Red curves show the injected signal; points show counts minus generator background.



All 20 independent Poisson draws appear in saved seed order (display toy 01 is stored ID 0). Every panel uses identical axes within this scenario and $\sqrt{n}$ count errors. “Z” is evaluated at the injection mass; “max” scans the declared nearby hypotheses, with its mass in MeV. An asterisk marks a grid-endpoint maximum.

75 MeV region: $10\% \times 10$

Injection at $m_0 = 78$ MeV, $\sigma_m = 2.263$ MeV; source $Z_f = 2.809$, requested $Z_{\text{target}} = 8.882$.



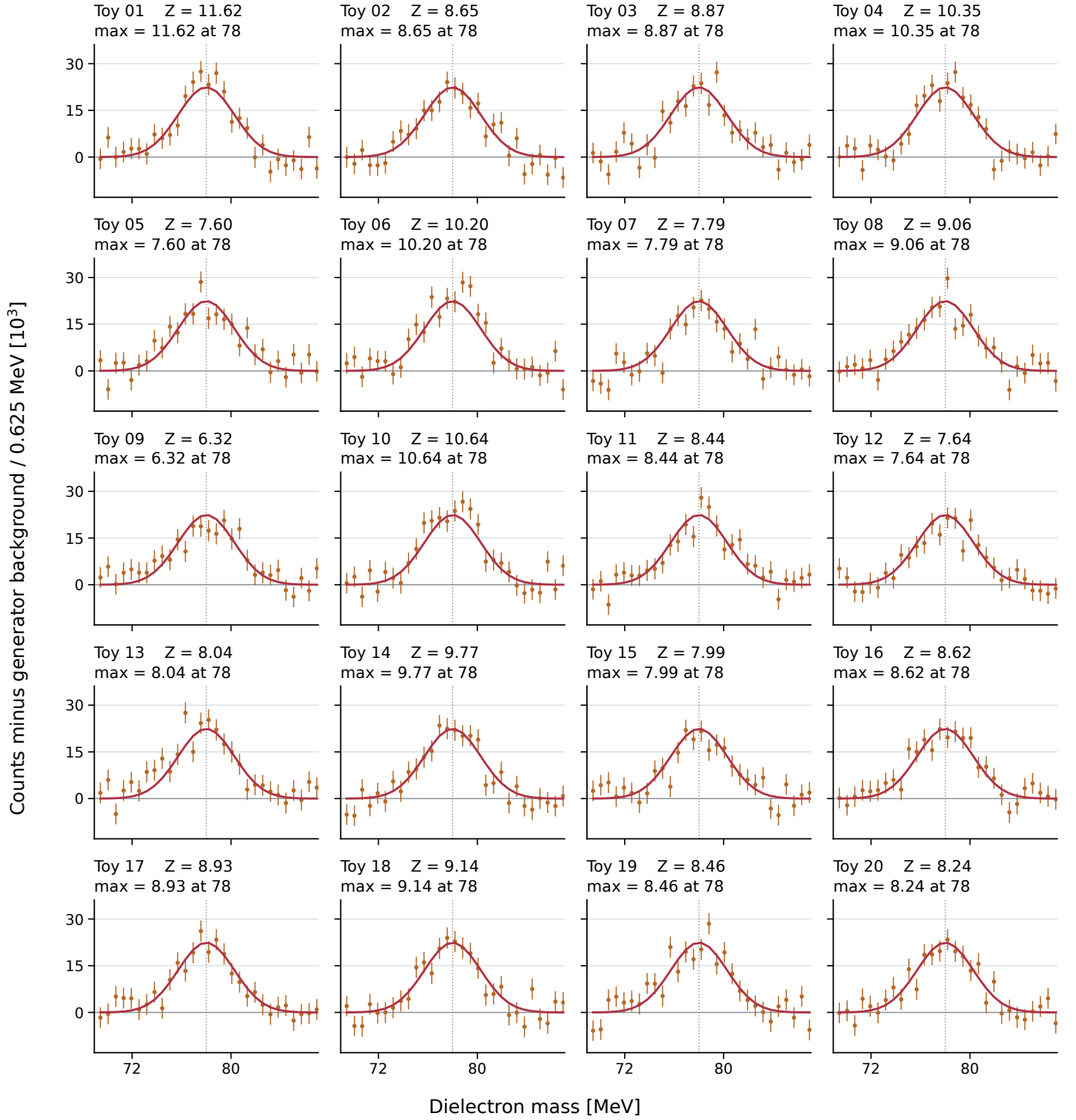
Shaded bands mark the $\pm 2.25\sigma_m$ extraction window. Observed error bars are $\sqrt{n}$; the dotted Asimov excess subtracts the profiled background in that window. The lower-right curves use fresh GP conditioning at every tested mass.

Yield-scaled amplitude a_{yield}	192,843	Matched amplitude a_{match}	204,300
Yield-scaled Asimov Z	8.395	Amplitude ratio R_a	1.059
Background-only Asimov Z	0.194	Matched Asimov $Z(m_0)$	8.882
Signal inside fit window	198,701	Background inside fit window	183,974,733
Fixed-mass median Z	8.64	Empirical 16–84% interval	[7.80, 10.18]
Largest fixed-mass Z	11.62	Largest nearby-maximum Z	11.62
Fixed-mass $Z \geq 3$	20/20	Fixed-mass $Z \geq 5$	20/20

The nearby scan covers 74–82 MeV in 1 MeV steps (9 hypotheses). Its median maximum is 8.64; 0/20 toy maxima lie at a grid endpoint. The matched Asimov maximum is 8.88 at 78 MeV.

75 MeV region: $10\% \times 10$: 20 realizations

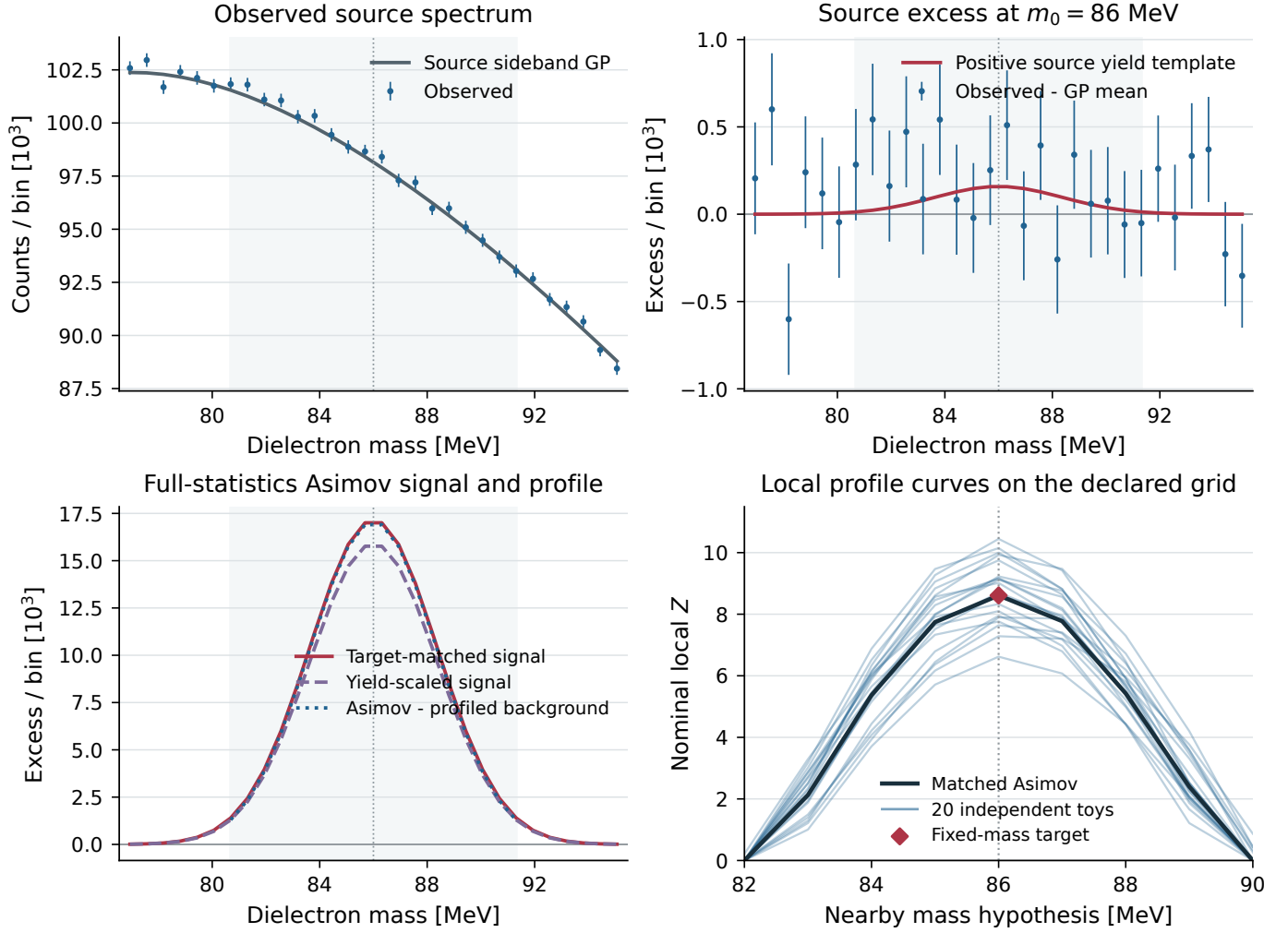
$10\% \times 10$; $m_0 = 78$ MeV; $Z_{\text{target}} = 8.88$. Red curves show the injected signal; points show counts minus generator background.



All 20 independent Poisson draws appear in saved seed order (display toy 01 is stored ID 0). Every panel uses identical axes within this scenario and $\sqrt{n}$ count errors. “Z” is evaluated at the injection mass; “max” scans the declared nearby hypotheses, with its mass in MeV. An asterisk marks a grid-endpoint maximum.

92 MeV region: $1\% \times 100$

Injection at $m_0 = 86$ MeV, $\sigma_m = 2.365$ MeV; source $Z_f = 0.861$, requested $Z_{\text{target}} = 8.610$. The source maximum is at the 86–99 MeV interval boundary.



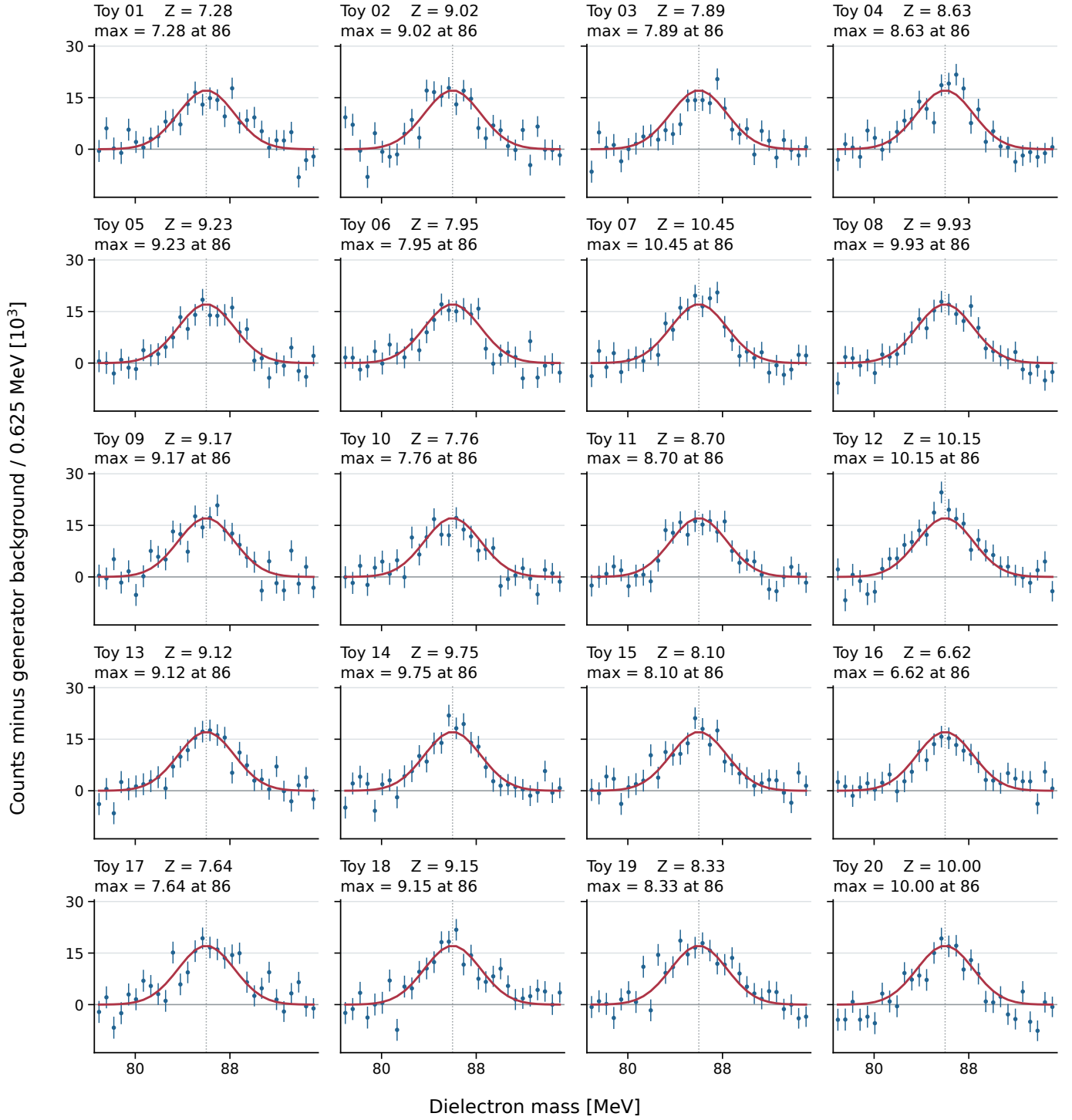
Shaded bands mark the $\pm 2.25\sigma_m$ extraction window. Observed error bars are $\sqrt{n}$; the dotted Asimov excess subtracts the profiled background in that window. The lower-right curves use fresh GP conditioning at every tested mass.

Yield-scaled amplitude a_{yield}	151,292	Matched amplitude a_{match}	163,160
Yield-scaled Asimov Z	7.990	Amplitude ratio R_a	1.078
Background-only Asimov Z	0.087	Matched Asimov $Z(m_0)$	8.610
Signal inside fit window	160,322	Background inside fit window	176,122,261
Fixed-mass median Z	8.86	Empirical 16–84% interval	[7.77, 9.92]
Largest fixed-mass Z	10.45	Largest nearby-maximum Z	10.45
Fixed-mass $Z \geq 3$	20/20	Fixed-mass $Z \geq 5$	20/20

The nearby scan covers 82–90 MeV in 1 MeV steps (9 hypotheses). Its median maximum is 8.86; 0/20 toy maxima lie at a grid endpoint. The matched Asimov maximum is 8.61 at 86 MeV.

92 MeV region: $1\% \times 100$: 20 realizations

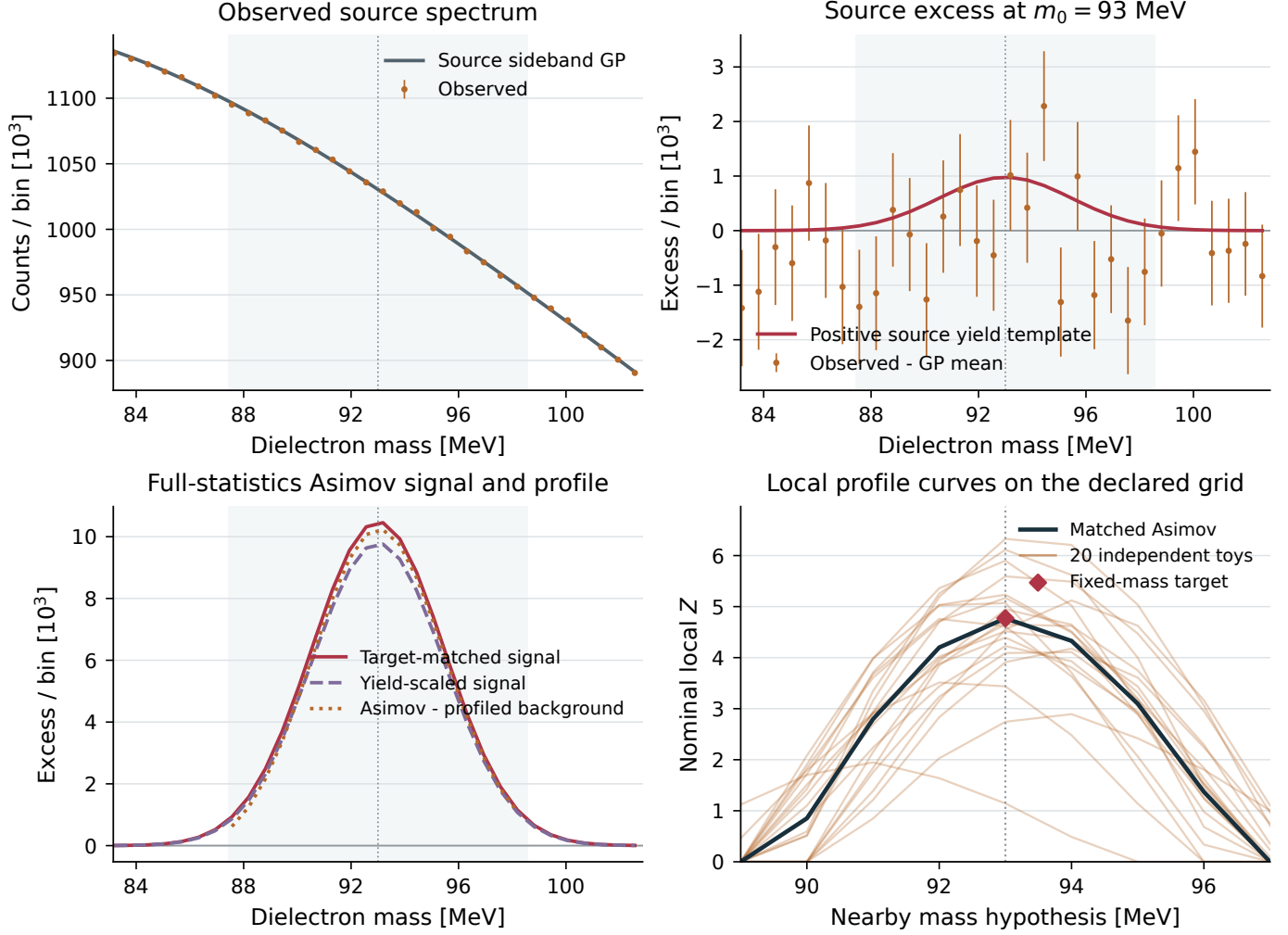
$1\% \times 100$; $m_0 = 86$ MeV; $Z_{\text{target}} = 8.61$. Red curves show the injected signal; points show counts minus generator background.



All 20 independent Poisson draws appear in saved seed order (display toy 01 is stored ID 0). Every panel uses identical axes within this scenario and $\sqrt{n}$ count errors. “Z” is evaluated at the injection mass; “max” scans the declared nearby hypotheses, with its mass in MeV. An asterisk marks a grid-endpoint maximum.

92 MeV region: $10\% \times 10$

Injection at $m_0 = 93$ MeV, $\sigma_m = 2.463$ MeV; source $Z_f = 1.509$, requested $Z_{\text{target}} = 4.771$.



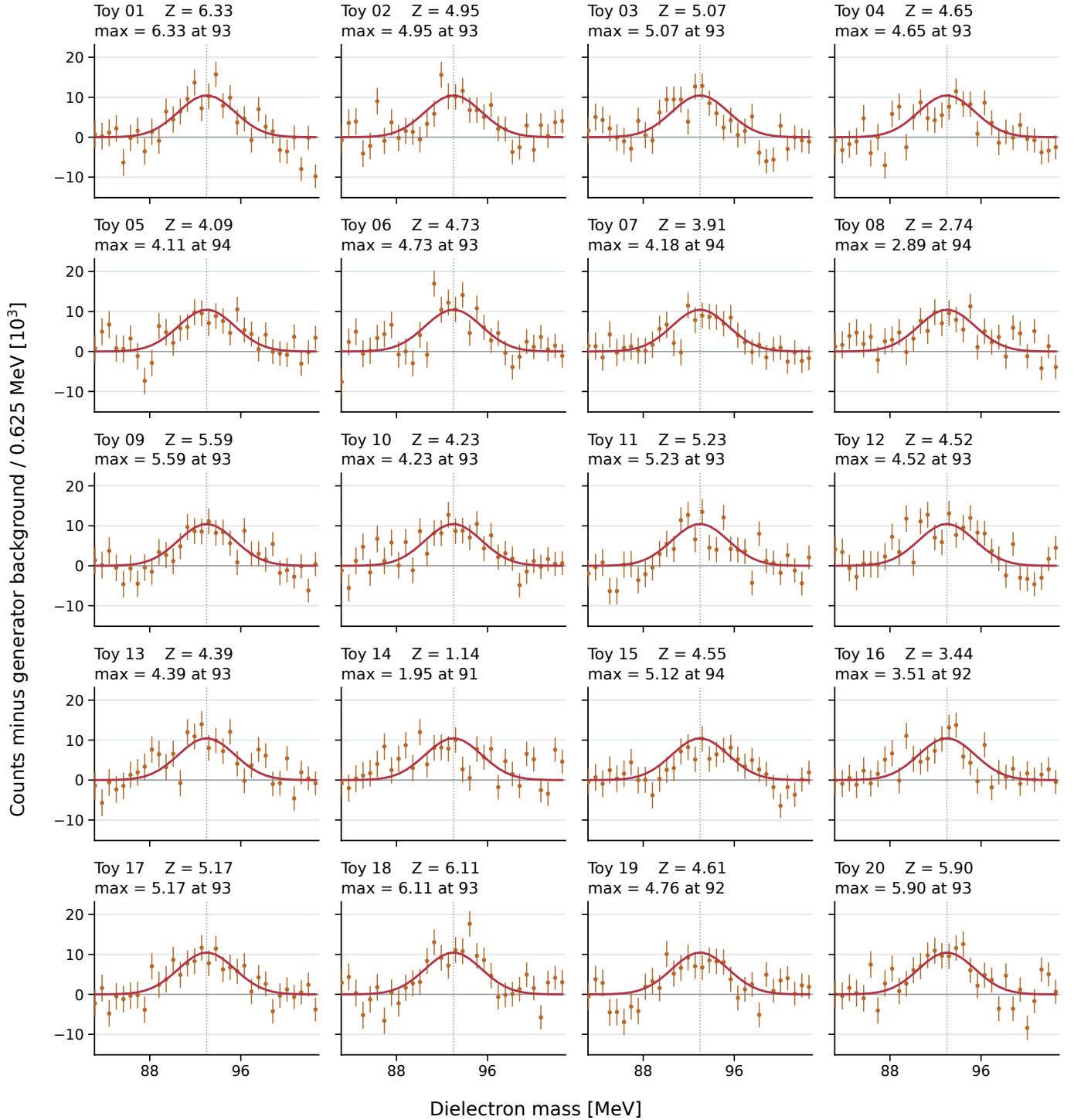
Shaded bands mark the $\pm 2.25\sigma_m$ extraction window. Observed error bars are $\sqrt{n}$; the dotted Asimov excess subtracts the profiled background in that window. The lower-right curves use fresh GP conditioning at every tested mass.

Yield-scaled amplitude a_{yield}	96,941	Matched amplitude a_{match}	103,832
Yield-scaled Asimov Z	4.454	Amplitude ratio R_a	1.071
Background-only Asimov Z	0.000	Matched Asimov $Z(m_0)$	4.771
Signal inside fit window	101,489	Background inside fit window	185,422,989
Fixed-mass median Z	4.63	Empirical 16–84% interval	[3.92, 5.58]
Largest fixed-mass Z	6.33	Largest nearby-maximum Z	6.33
Fixed-mass $Z \geq 3$	18/20	Fixed-mass $Z \geq 5$	7/20

The nearby scan covers 89–97 MeV in 1 MeV steps (9 hypotheses). Its median maximum is 4.74; 0/20 toy maxima lie at a grid endpoint. The matched Asimov maximum is 4.77 at 93 MeV.

92 MeV region: $10\% \times 10$: 20 realizations

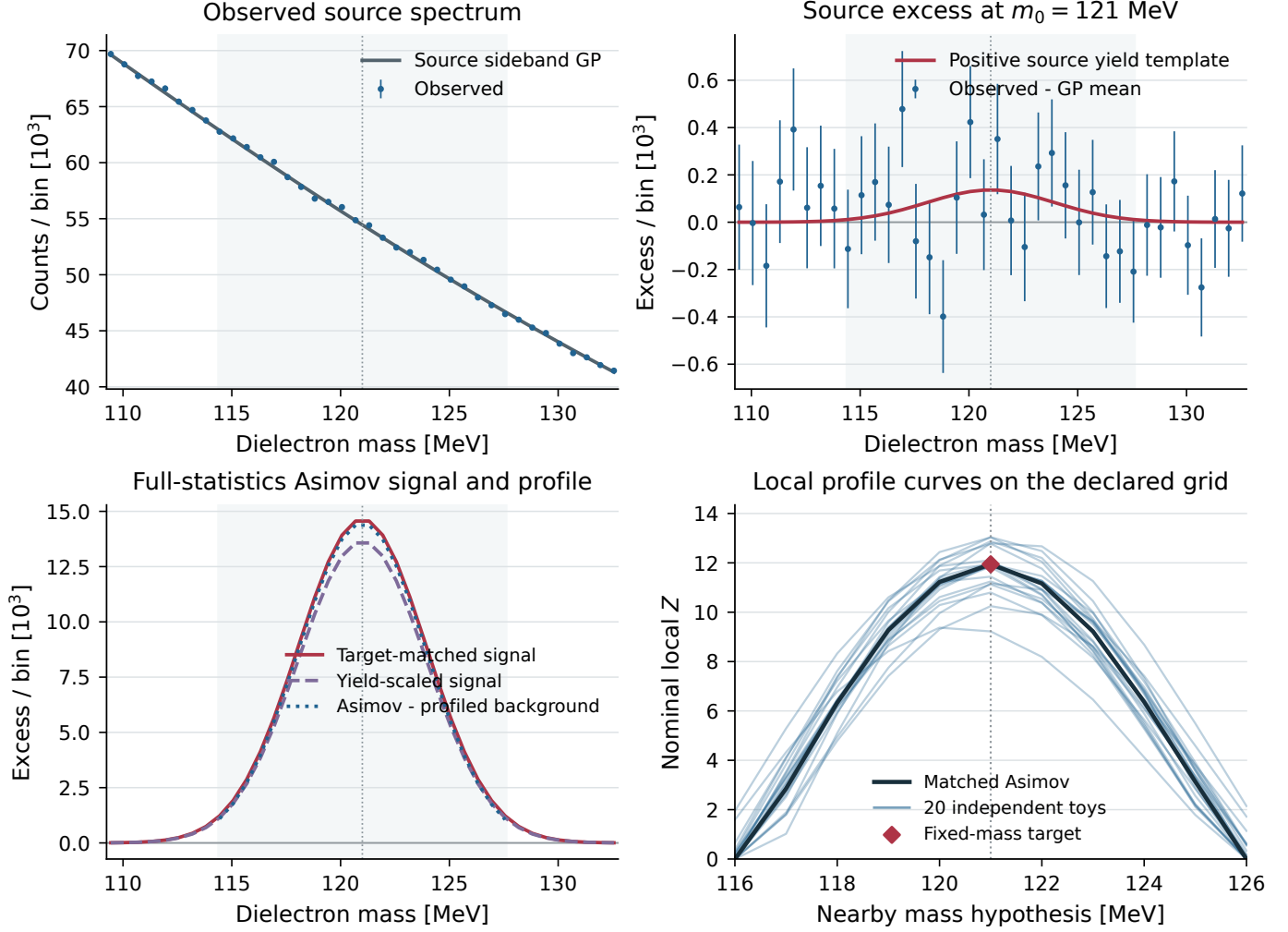
$10\% \times 10$; $m_0 = 93$ MeV; $Z_{\text{target}} = 4.77$. Red curves show the injected signal; points show counts minus generator background.



All 20 independent Poisson draws appear in saved seed order (display toy 01 is stored ID 0). Every panel uses identical axes within this scenario and $\sqrt{n}$ count errors. “Z” is evaluated at the injection mass; “max” scans the declared nearby hypotheses, with its mass in MeV. An asterisk marks a grid-endpoint maximum.

120 MeV region: $1\% \times 100$

Injection at $m_0 = 121$ MeV, $\sigma_m = 2.939$ MeV; source $Z_f = 1.194$, requested $Z_{\text{target}} = 11.943$.



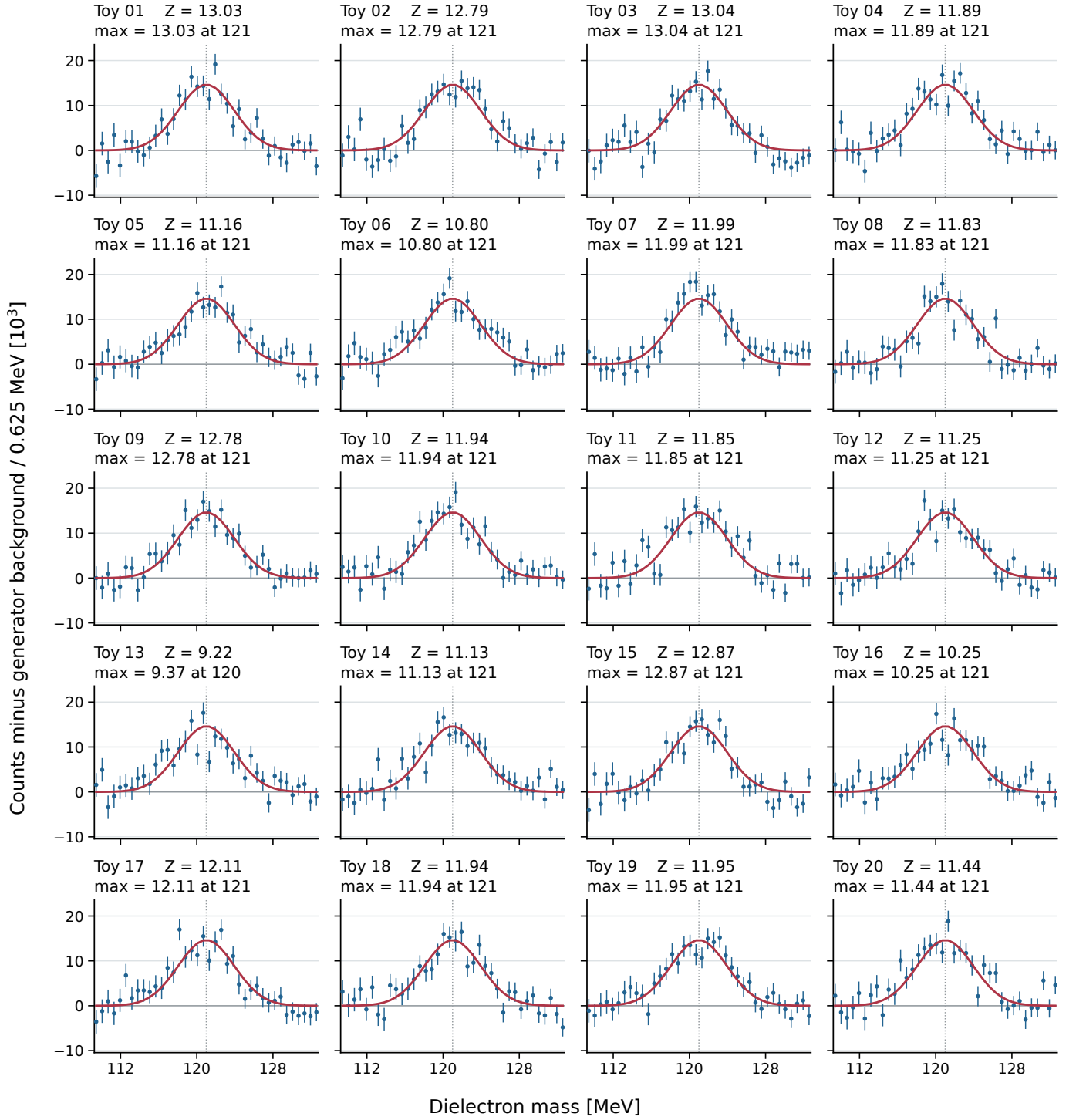
Shaded bands mark the $\pm 2.25\sigma_m$ extraction window. Observed error bars are $\sqrt{n}$; the dotted Asimov excess subtracts the profiled background in that window. The lower-right curves use fresh GP conditioning at every tested mass.

Yield-scaled amplitude a_{yield}	161,177	Matched amplitude a_{match}	172,949
Yield-scaled Asimov Z	11.131	Amplitude ratio R_a	1.073
Background-only Asimov Z	0.003	Matched Asimov $Z(m_0)$	11.943
Signal inside fit window	169,606	Background inside fit window	120,058,900
Fixed-mass median Z	11.91	Empirical 16–84% interval	[11.13, 12.79]
Largest fixed-mass Z	13.04	Largest nearby-maximum Z	13.04
Fixed-mass $Z \geq 3$	20/20	Fixed-mass $Z \geq 5$	20/20

The nearby scan covers 116–126 MeV in 1 MeV steps (11 hypotheses). Its median maximum is 11.91; 0/20 toy maxima lie at a grid endpoint. The matched Asimov maximum is 11.94 at 121 MeV.

120 MeV region: $1\% \times 100$: 20 realizations

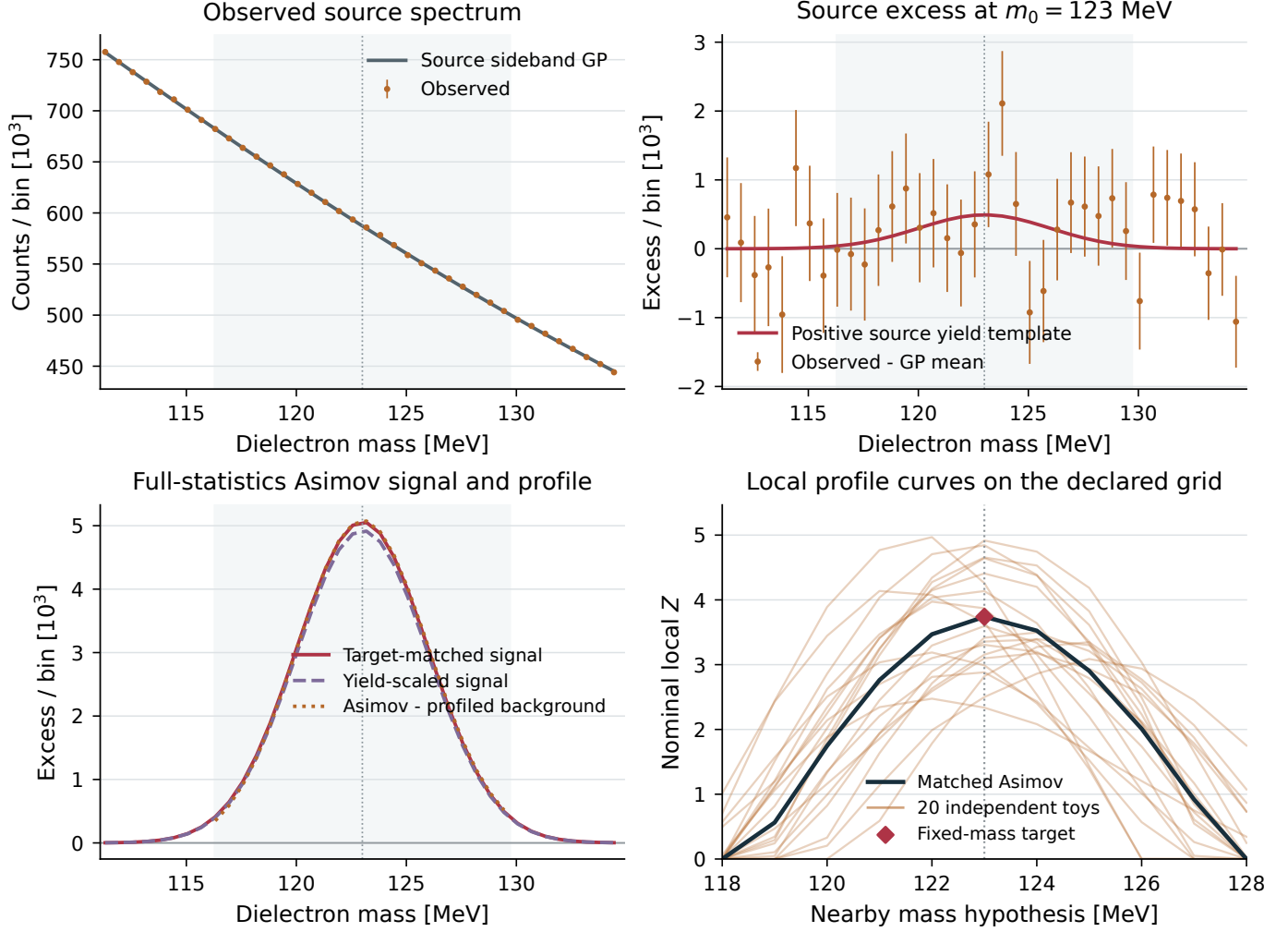
$1\% \times 100$; $m_0 = 121$ MeV; $Z_{\text{target}} = 11.94$. Red curves show the injected signal; points show counts minus generator background.



All 20 independent Poisson draws appear in saved seed order (display toy 01 is stored ID 0). Every panel uses identical axes within this scenario and $\sqrt{n}$ count errors. “ Z ” is evaluated at the injection mass; “max” scans the declared nearby hypotheses, with its mass in MeV. An asterisk marks a grid-endpoint maximum.

120 MeV region: $10\% \times 10$

Injection at $m_0 = 123$ MeV, $\sigma_m = 2.978$ MeV; source $Z_f = 1.182$, requested $Z_{\text{target}} = 3.739$.



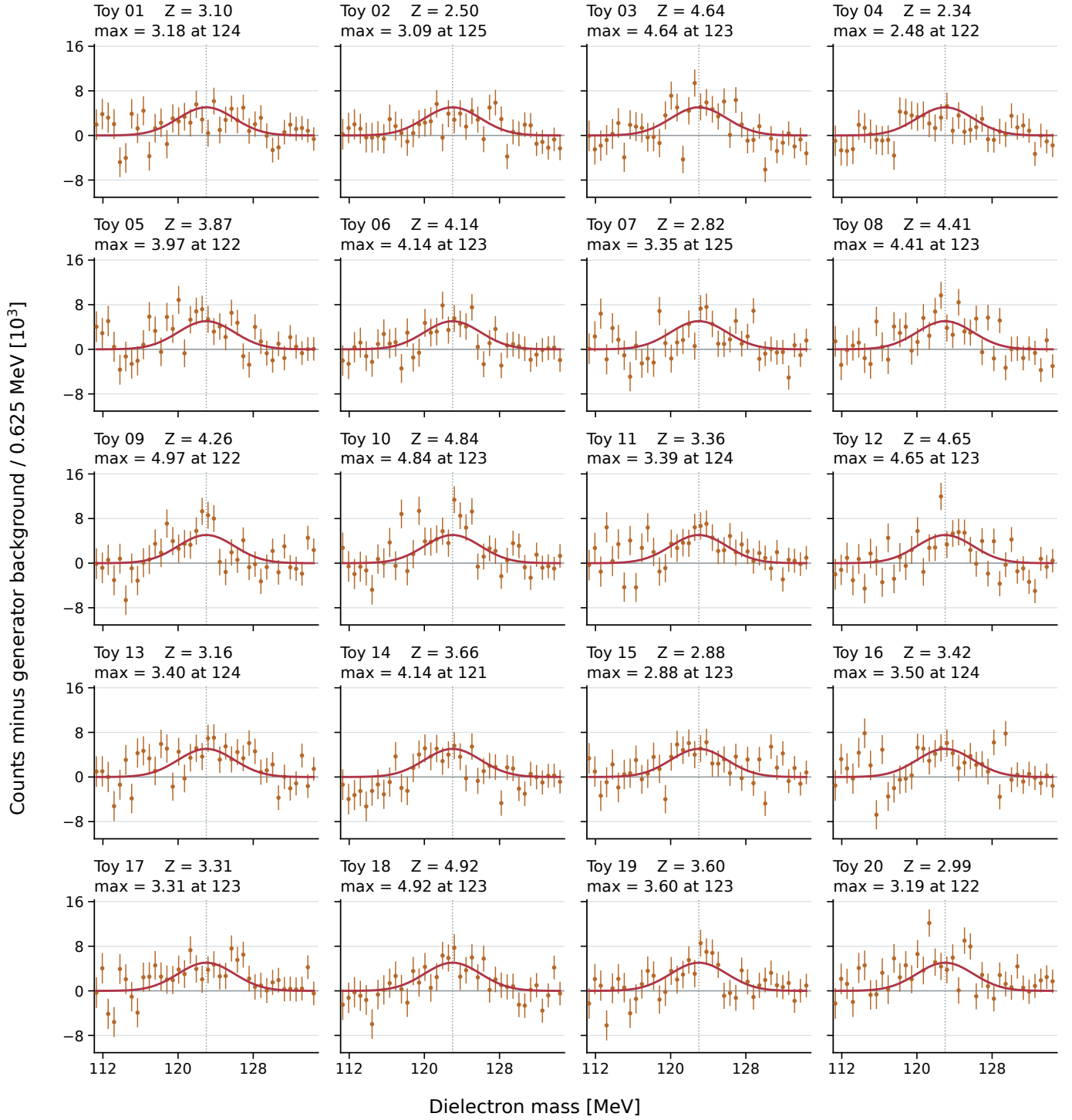
Shaded bands mark the $\pm 2.25\sigma_m$ extraction window. Observed error bars are $\sqrt{n}$; the dotted Asimov excess subtracts the profiled background in that window. The lower-right curves use fresh GP conditioning at every tested mass.

Yield-scaled amplitude a_{yield}	58,924	Matched amplitude a_{match}	60,570
Yield-scaled Asimov Z	3.640	Amplitude ratio R_a	1.028
Background-only Asimov Z	0.099	Matched Asimov $Z(m_0)$	3.739
Signal inside fit window	59,292	Background inside fit window	129,904,521
Fixed-mass median Z	3.51	Empirical 16–84% interval	[2.89, 4.63]
Largest fixed-mass Z	4.92	Largest nearby-maximum Z	4.97
Fixed-mass $Z \geq 3$	15/20	Fixed-mass $Z \geq 5$	0/20

The nearby scan covers 118–128 MeV in 1 MeV steps (11 hypotheses). Its median maximum is 3.55; 0/20 toy maxima lie at a grid endpoint. The matched Asimov maximum is 3.74 at 123 MeV.

120 MeV region: $10\% \times 10$: 20 realizations

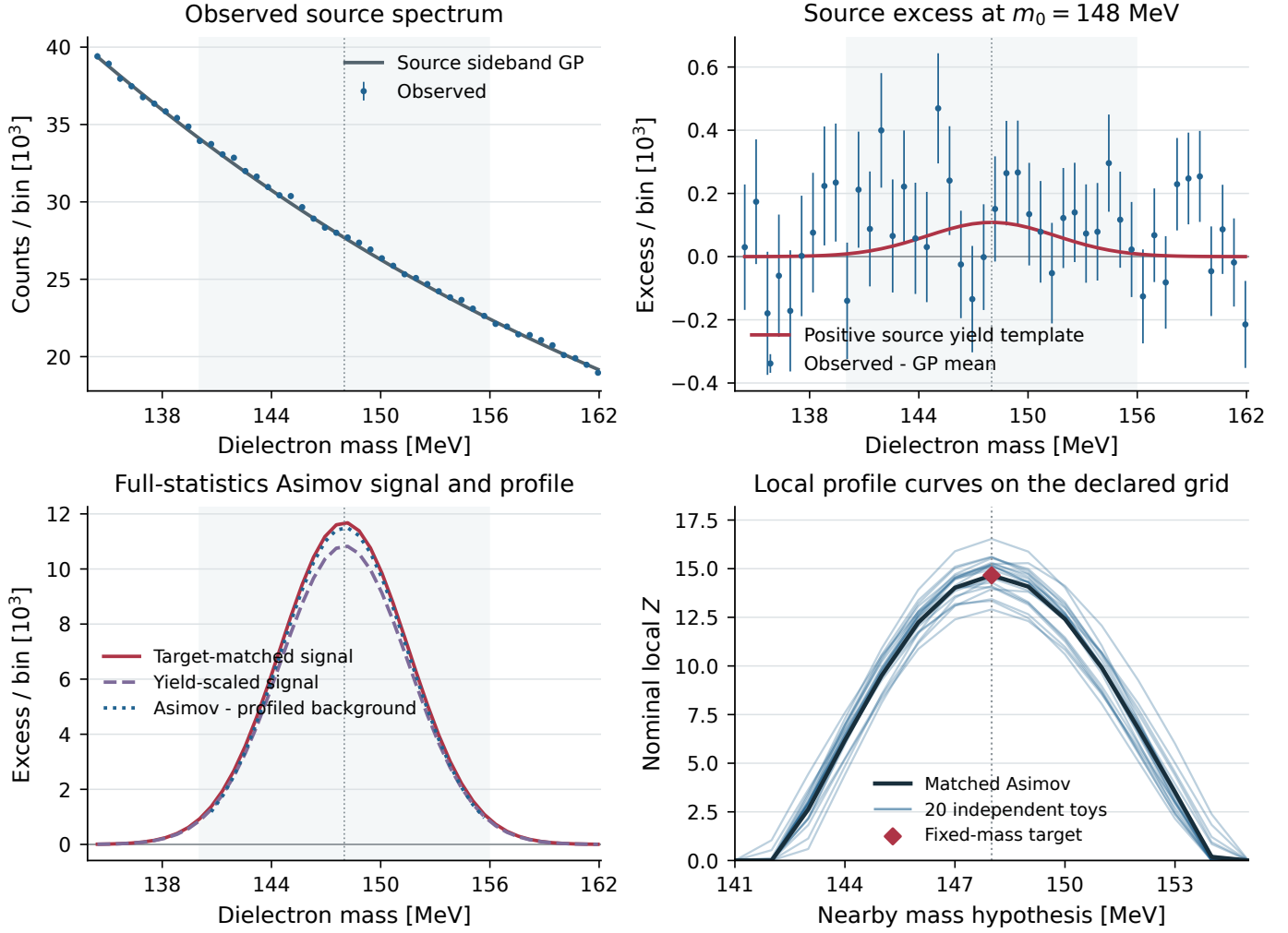
$10\% \times 10$; $m_0 = 123$ MeV; $Z_{\text{target}} = 3.74$. Red curves show the injected signal; points show counts minus generator background.



All 20 independent Poisson draws appear in saved seed order (display toy 01 is stored ID 0). Every panel uses identical axes within this scenario and $\sqrt{n}$ count errors. “Z” is evaluated at the injection mass; “max” scans the declared nearby hypotheses, with its mass in MeV. An asterisk marks a grid-endpoint maximum.

160 MeV region: $1\% \times 100$

Injection at $m_0 = 148$ MeV, $\sigma_m = 3.526$ MeV; source $Z_f = 1.466$, requested $Z_{\text{target}} = 14.659$. The source maximum is at the 148–177 MeV interval boundary.



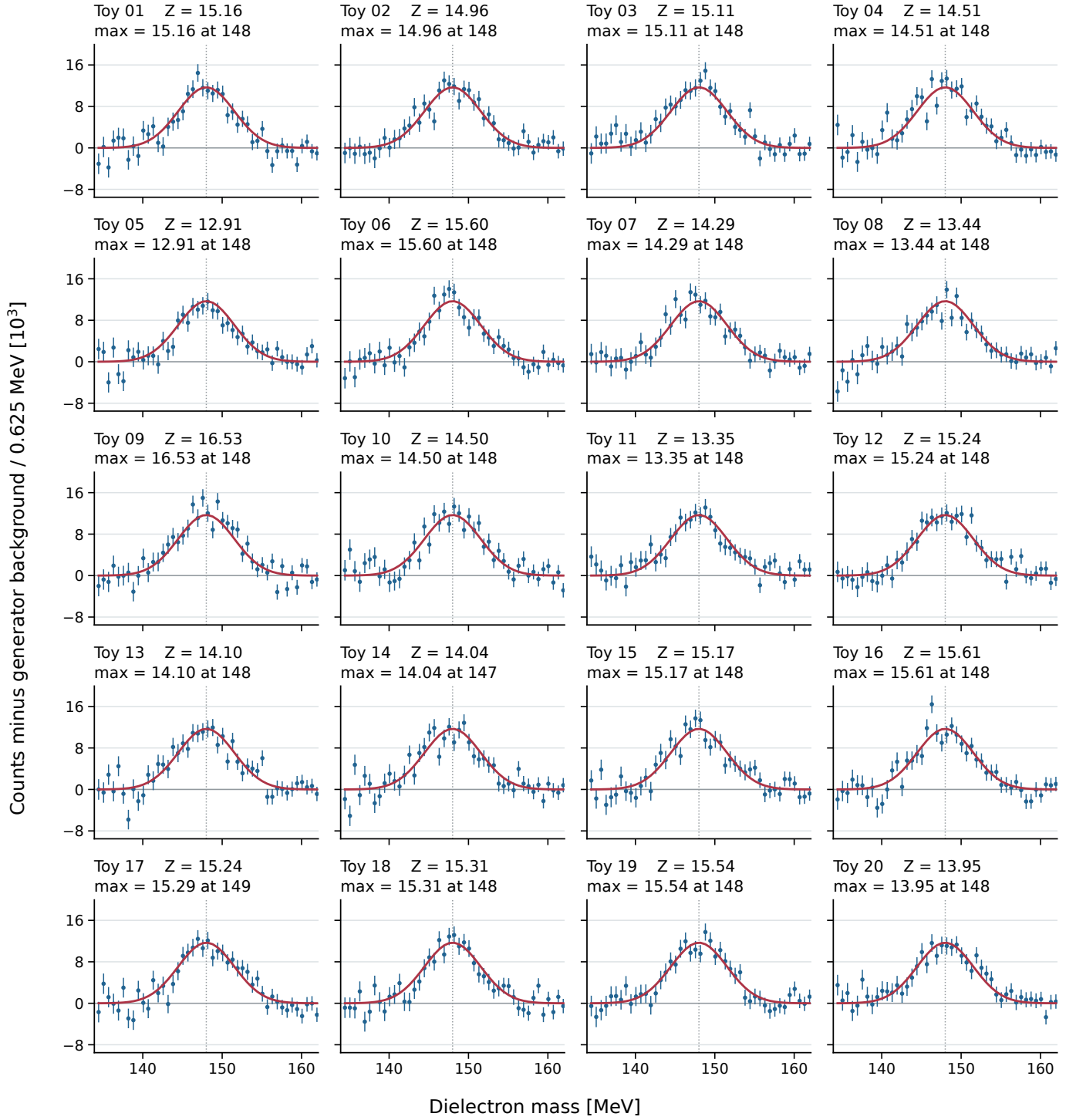
Shaded bands mark the $\pm 2.25\sigma_m$ extraction window. Observed error bars are $\sqrt{n}$; the dotted Asimov excess subtracts the profiled background in that window. The lower-right curves use fresh GP conditioning at every tested mass.

Yield-scaled amplitude a_{yield}	153,441	Matched amplitude a_{match}	165,553
Yield-scaled Asimov Z	13.590	Amplitude ratio R_a	1.079
Background-only Asimov Z	0.047	Matched Asimov $Z(m_0)$	14.659
Signal inside fit window	161,097	Background inside fit window	69,344,428
Fixed-mass median Z	15.03	Empirical 16–84% interval	[13.96, 15.53]
Largest fixed-mass Z	16.53	Largest nearby-maximum Z	16.53
Fixed-mass $Z \geq 3$	20/20	Fixed-mass $Z \geq 5$	20/20

The nearby scan covers 141–155 MeV in 1 MeV steps (15 hypotheses). Its median maximum is 15.03; 0/20 toy maxima lie at a grid endpoint. The matched Asimov maximum is 14.66 at 148 MeV.

160 MeV region: $1\% \times 100$: 20 realizations

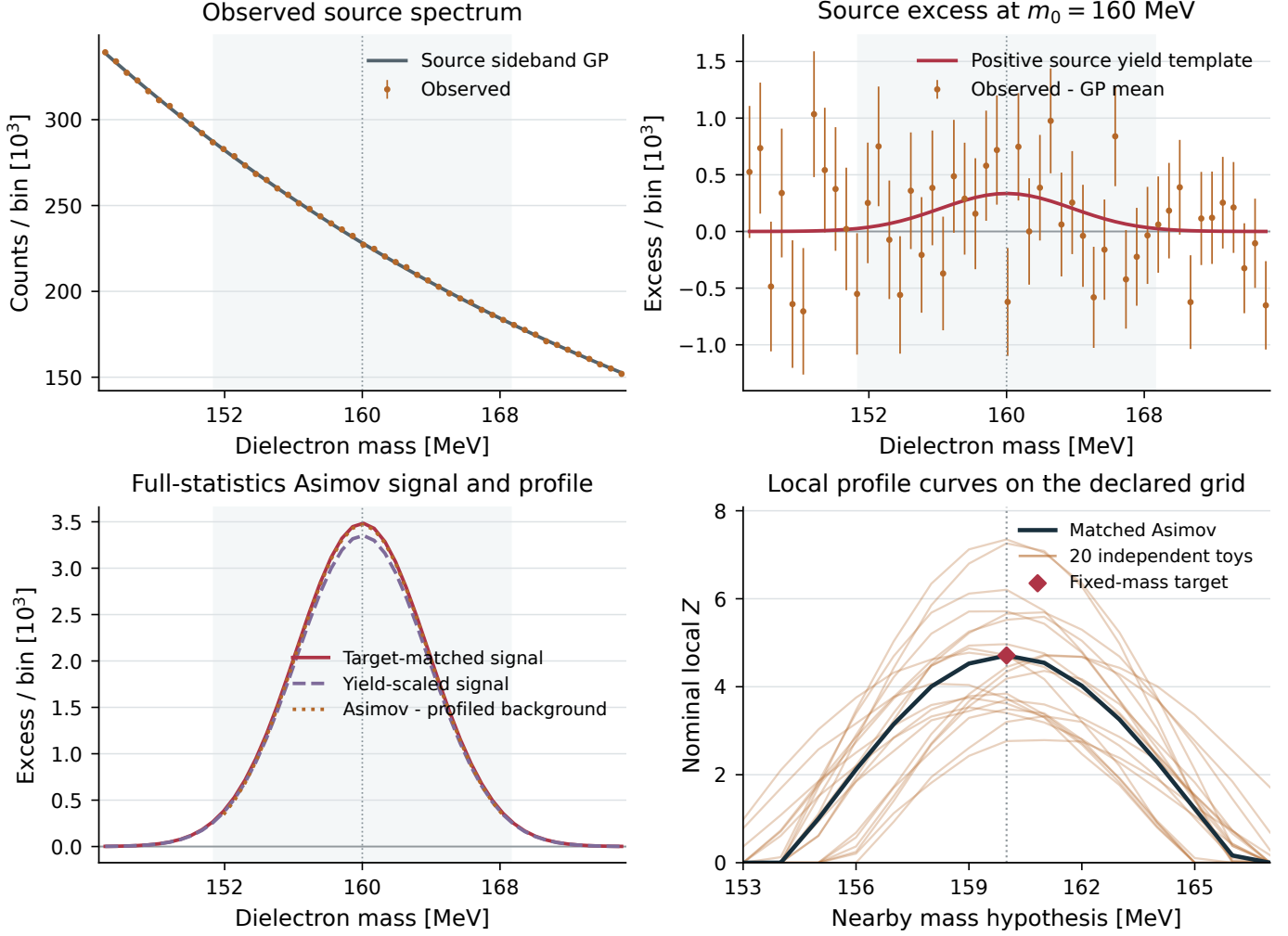
$1\% \times 100$; $m_0 = 148$ MeV; $Z_{\text{target}} = 14.66$. Red curves show the injected signal; points show counts minus generator background.



All 20 independent Poisson draws appear in saved seed order (display toy 01 is stored ID 0). Every panel uses identical axes within this scenario and $\sqrt{n}$ count errors. “Z” is evaluated at the injection mass; “max” scans the declared nearby hypotheses, with its mass in MeV. An asterisk marks a grid-endpoint maximum.

160 MeV region: $10\% \times 10$

Injection at $m_0 = 160$ MeV, $\sigma_m = 3.827$ MeV; source $Z_f = 1.490$, requested $Z_{\text{target}} = 4.710$.



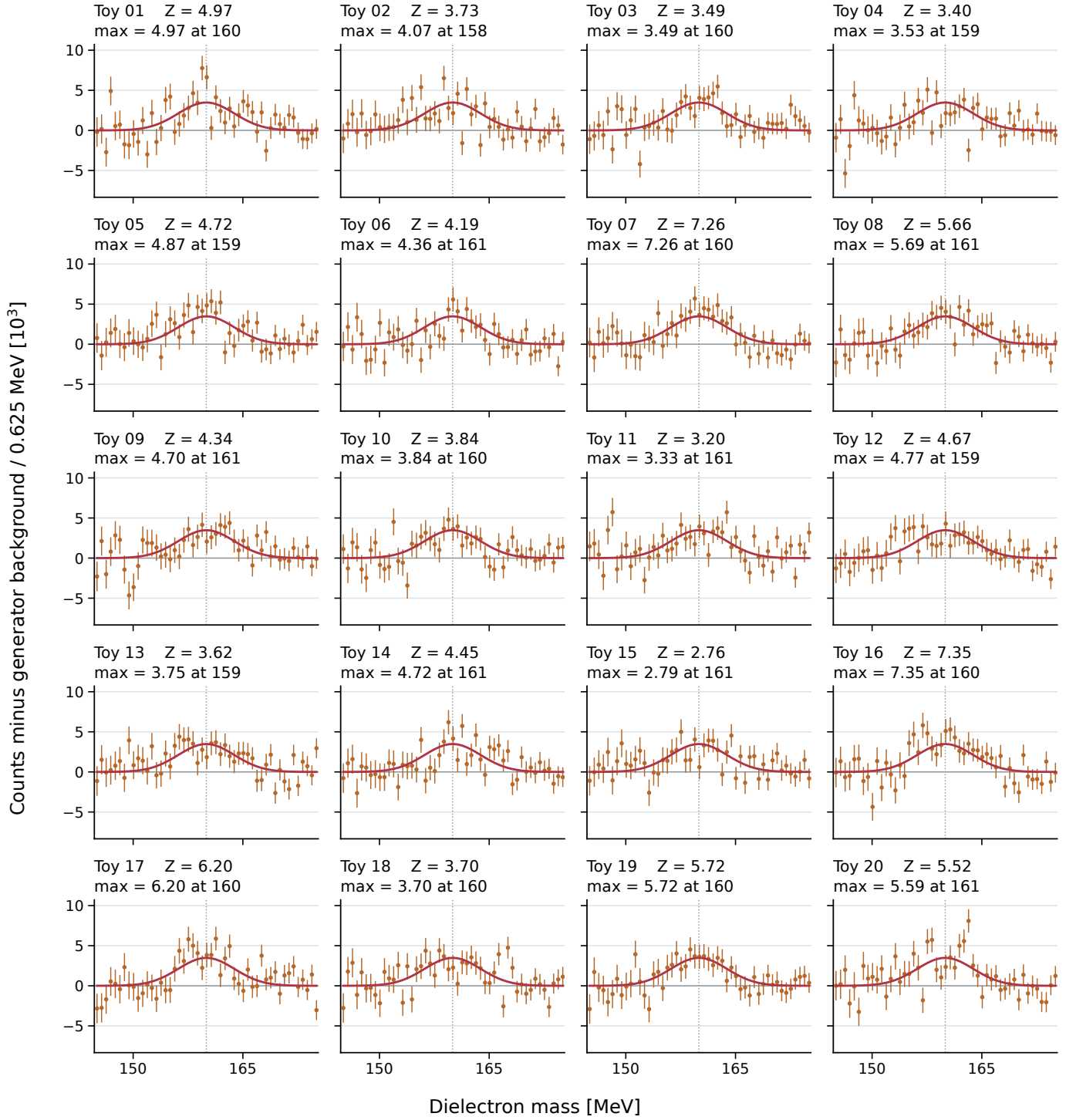
Shaded bands mark the $\pm 2.25\sigma_m$ extraction window. Observed error bars are $\sqrt{n}$; the dotted Asimov excess subtracts the profiled background in that window. The lower-right curves use fresh GP conditioning at every tested mass.

Yield-scaled amplitude a_{yield}	51,512	Matched amplitude a_{match}	53,539
Yield-scaled Asimov Z	4.536	Amplitude ratio R_a	1.039
Background-only Asimov Z	0.106	Matched Asimov $Z(m_0)$	4.710
Signal inside fit window	52,068	Background inside fit window	62,008,303
Fixed-mass median Z	4.39	Empirical 16–84% interval	[3.50, 5.71]
Largest fixed-mass Z	7.35	Largest nearby-maximum Z	7.35
Fixed-mass $Z \geq 3$	19/20	Fixed-mass $Z \geq 5$	6/20

The nearby scan covers 153–167 MeV in 1 MeV steps (15 hypotheses). Its median maximum is 4.71; 0/20 toy maxima lie at a grid endpoint. The matched Asimov maximum is 4.71 at 160 MeV.

160 MeV region: $10\% \times 10$: 20 realizations

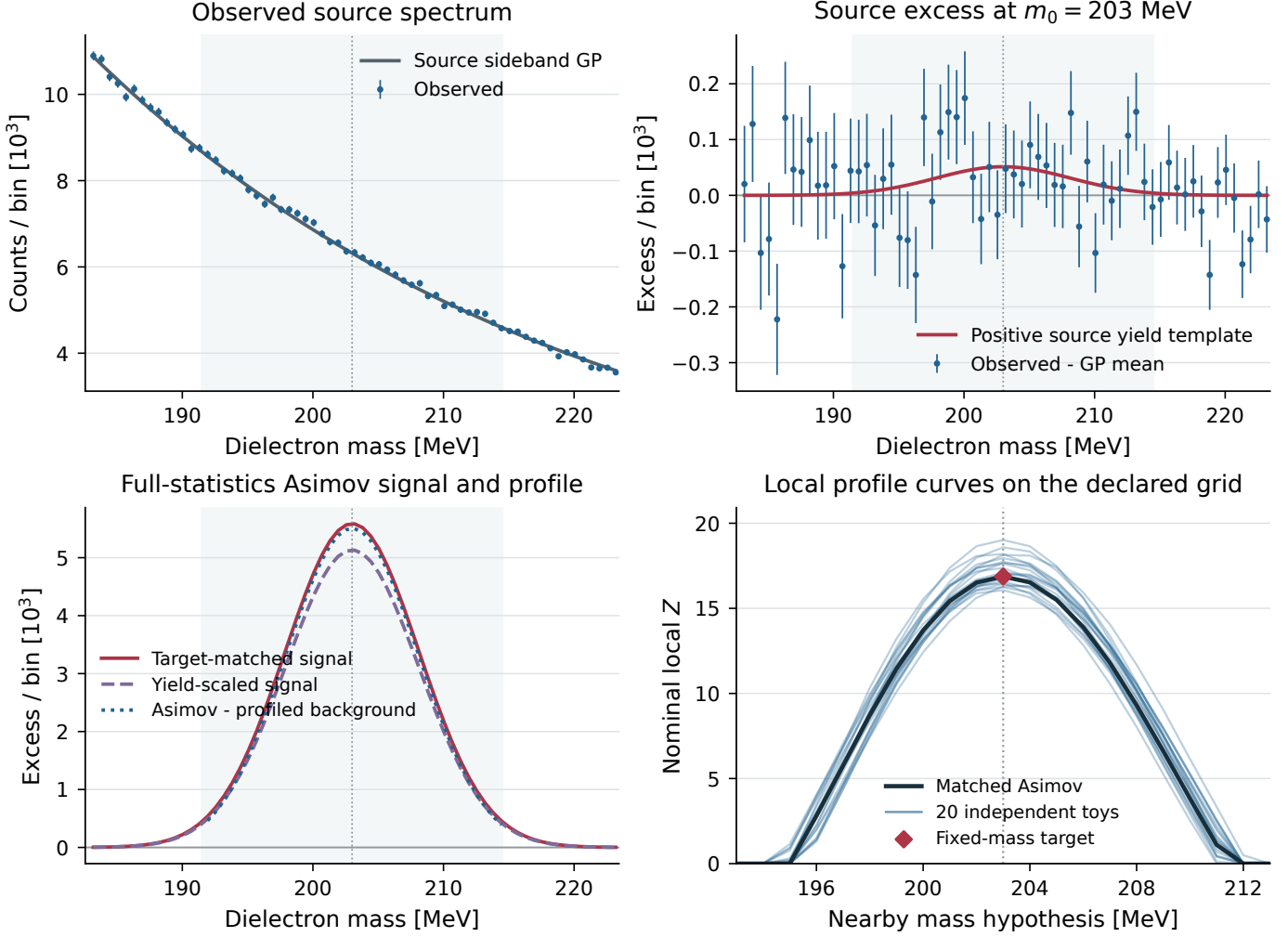
$10\% \times 10$; $m_0 = 160$ MeV; $Z_{\text{target}} = 4.71$. Red curves show the injected signal; points show counts minus generator background.



All 20 independent Poisson draws appear in saved seed order (display toy 01 is stored ID 0). Every panel uses identical axes within this scenario and $\sqrt{n}$ count errors. “ Z ” is evaluated at the injection mass; “max” scans the declared nearby hypotheses, with its mass in MeV. An asterisk marks a grid-endpoint maximum.

210 MeV region: $1\% \times 100$

Injection at $m_0 = 203$ MeV, $\sigma_m = 5.108$ MeV; source $Z_f = 1.687$, requested $Z_{\text{target}} = 16.870$.



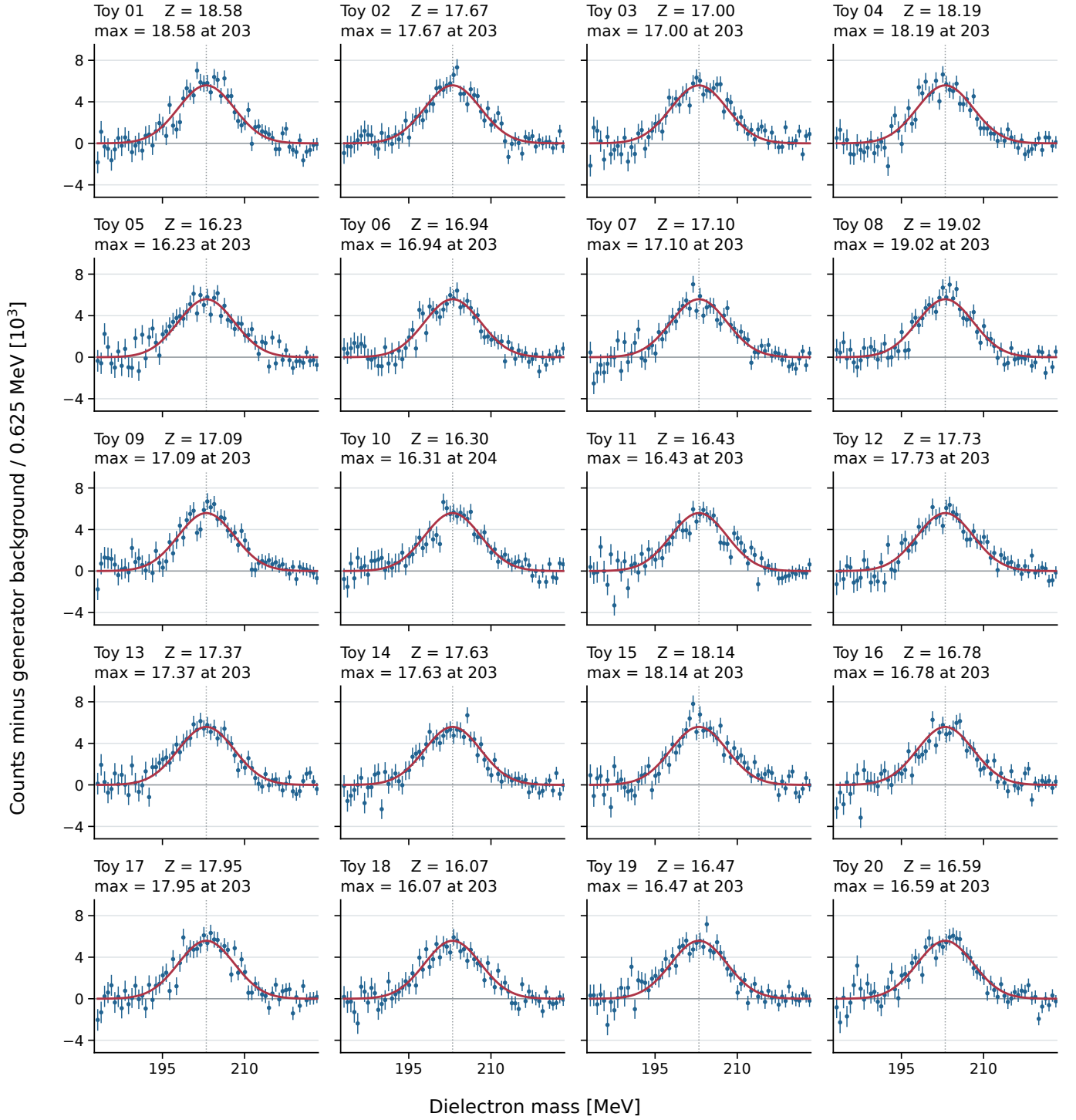
Shaded bands mark the $\pm 2.25\sigma_m$ extraction window. Observed error bars are $\sqrt{n}$; the dotted Asimov excess subtracts the profiled background in that window. The lower-right curves use fresh GP conditioning at every tested mass.

Yield-scaled amplitude a_{yield}	105,226	Matched amplitude a_{match}	114,554
Yield-scaled Asimov Z	15.503	Amplitude ratio R_a	1.089
Background-only Asimov Z	0.060	Matched Asimov $Z(m_0)$	16.870
Signal inside fit window	111,840	Background inside fit window	23,647,831
Fixed-mass median Z	17.09	Empirical 16–84% interval	[16.43, 18.13]
Largest fixed-mass Z	19.02	Largest nearby-maximum Z	19.02
Fixed-mass $Z \geq 3$	20/20	Fixed-mass $Z \geq 5$	20/20

The nearby scan covers 193–213 MeV in 1 MeV steps (21 hypotheses). Its median maximum is 17.09; 0/20 toy maxima lie at a grid endpoint. The matched Asimov maximum is 16.87 at 203 MeV.

210 MeV region: $1\% \times 100$: 20 realizations

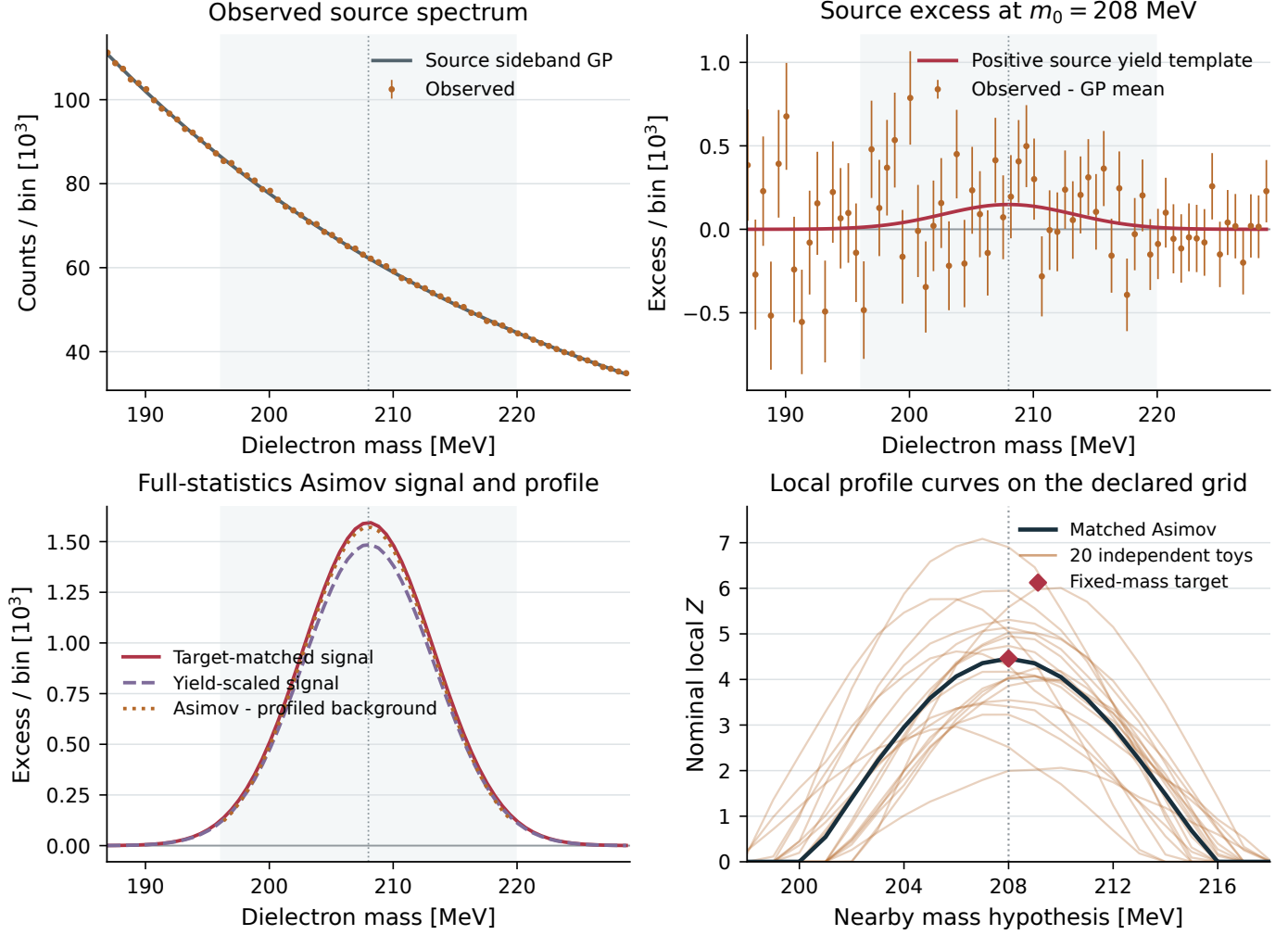
$1\% \times 100$; $m_0 = 203$ MeV; $Z_{\text{target}} = 16.87$. Red curves show the injected signal; points show counts minus generator background.



All 20 independent Poisson draws appear in saved seed order (display toy 01 is stored ID 0). Every panel uses identical axes within this scenario and $\sqrt{n}$ count errors. “Z” is evaluated at the injection mass; “max” scans the declared nearby hypotheses, with its mass in MeV. An asterisk marks a grid-endpoint maximum.

210 MeV region: $10\% \times 10$

Injection at $m_0 = 208$ MeV, $\sigma_m = 5.278$ MeV; source $Z_f = 1.410$, requested $Z_{\text{target}} = 4.459$.



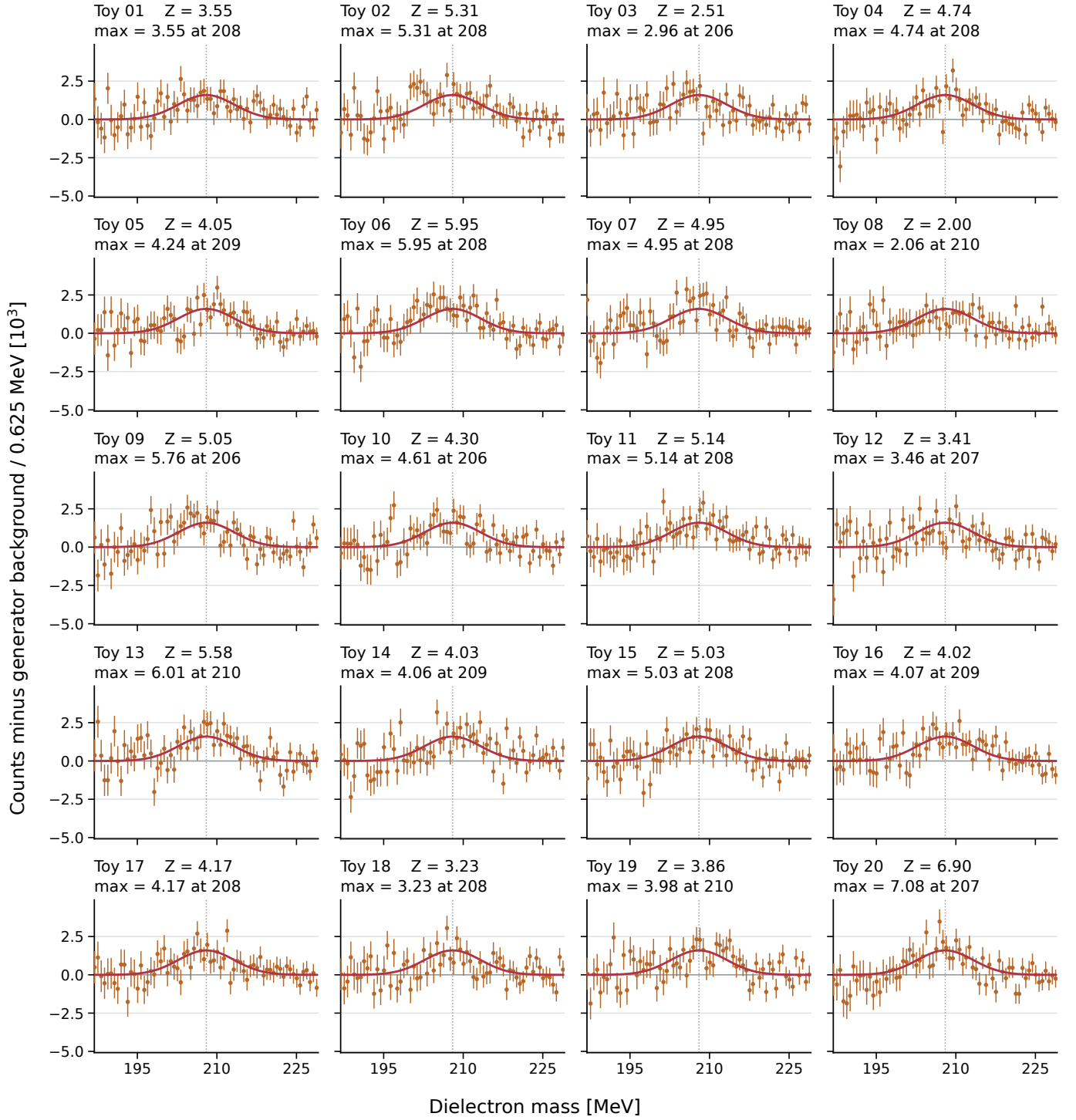
Shaded bands mark the $\pm 2.25\sigma_m$ extraction window. Observed error bars are $\sqrt{n}$; the dotted Asimov excess subtracts the profiled background in that window. The lower-right curves use fresh GP conditioning at every tested mass.

Yield-scaled amplitude a_{yield}	31,470	Matched amplitude a_{match}	33,765
Yield-scaled Asimov Z	4.160	Amplitude ratio R_a	1.073
Background-only Asimov Z	0.048	Matched Asimov $Z(m_0)$	4.459
Signal inside fit window	32,938	Background inside fit window	24,156,434
Fixed-mass median Z	4.23	Empirical 16–84% interval	[3.41, 5.30]
Largest fixed-mass Z	6.90	Largest nearby-maximum Z	7.08
Fixed-mass $Z \geq 3$	18/20	Fixed-mass $Z \geq 5$	7/20

The nearby scan covers 198–218 MeV in 1 MeV steps (21 hypotheses). Its median maximum is 4.43; 0/20 toy maxima lie at a grid endpoint. The matched Asimov maximum is 4.46 at 208 MeV.

210 MeV region: $10\% \times 10$: 20 realizations

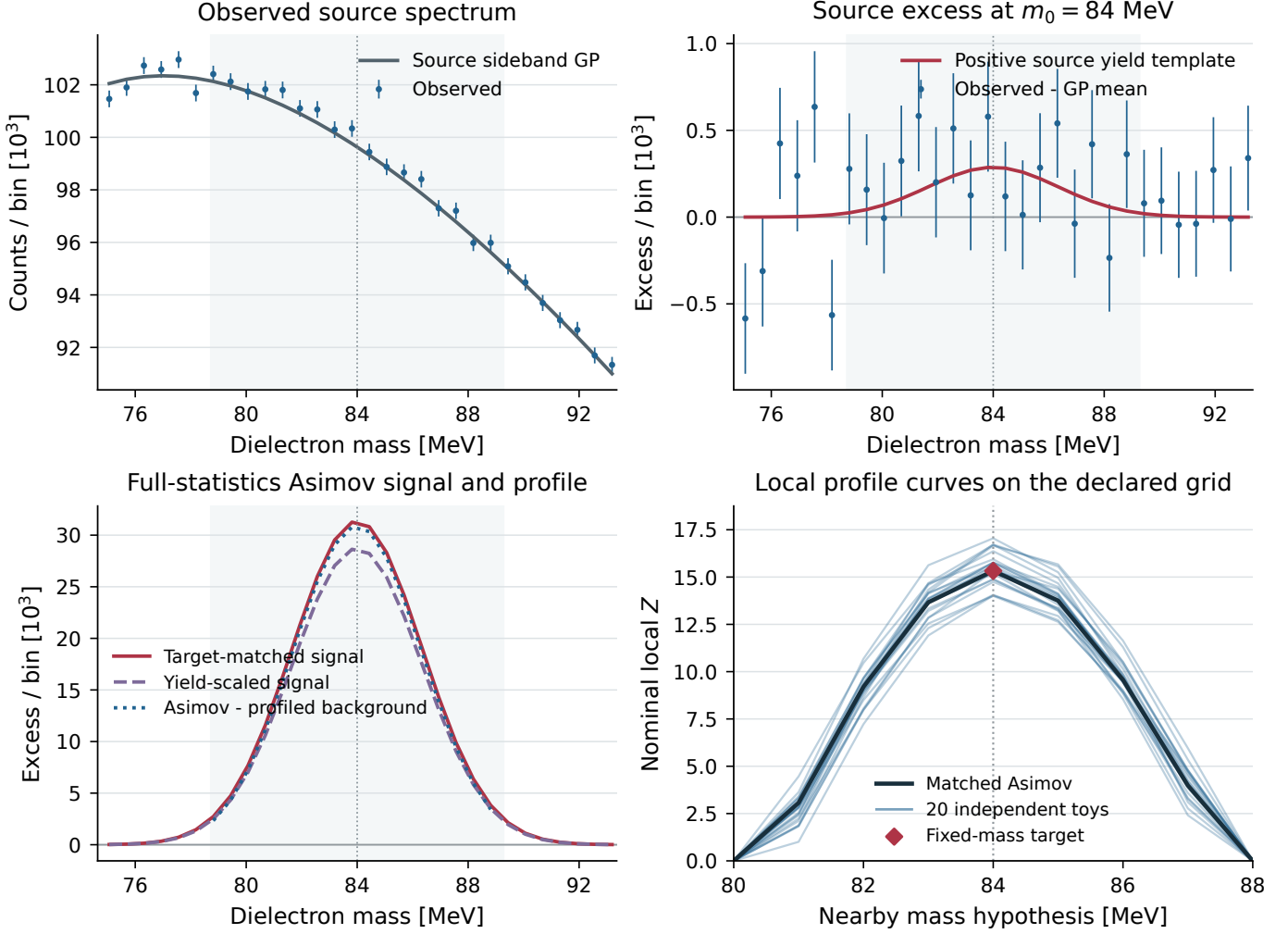
$10\% \times 10$; $m_0 = 208$ MeV; $Z_{\text{target}} = 4.46$. Red curves show the injected signal; points show counts minus generator background.



All 20 independent Poisson draws appear in saved seed order (display toy 01 is stored ID 0). Every panel uses identical axes within this scenario and $\sqrt{n}$ count errors. “Z” is evaluated at the injection mass; “max” scans the declared nearby hypotheses, with its mass in MeV. An asterisk marks a grid-endpoint maximum.

Additional 1% peak: 84 MeV

Injection at $m_0 = 84$ MeV, $\sigma_m = 2.339$ MeV; source $Z_f = 1.533$, requested $Z_{\text{target}} = 15.326$.



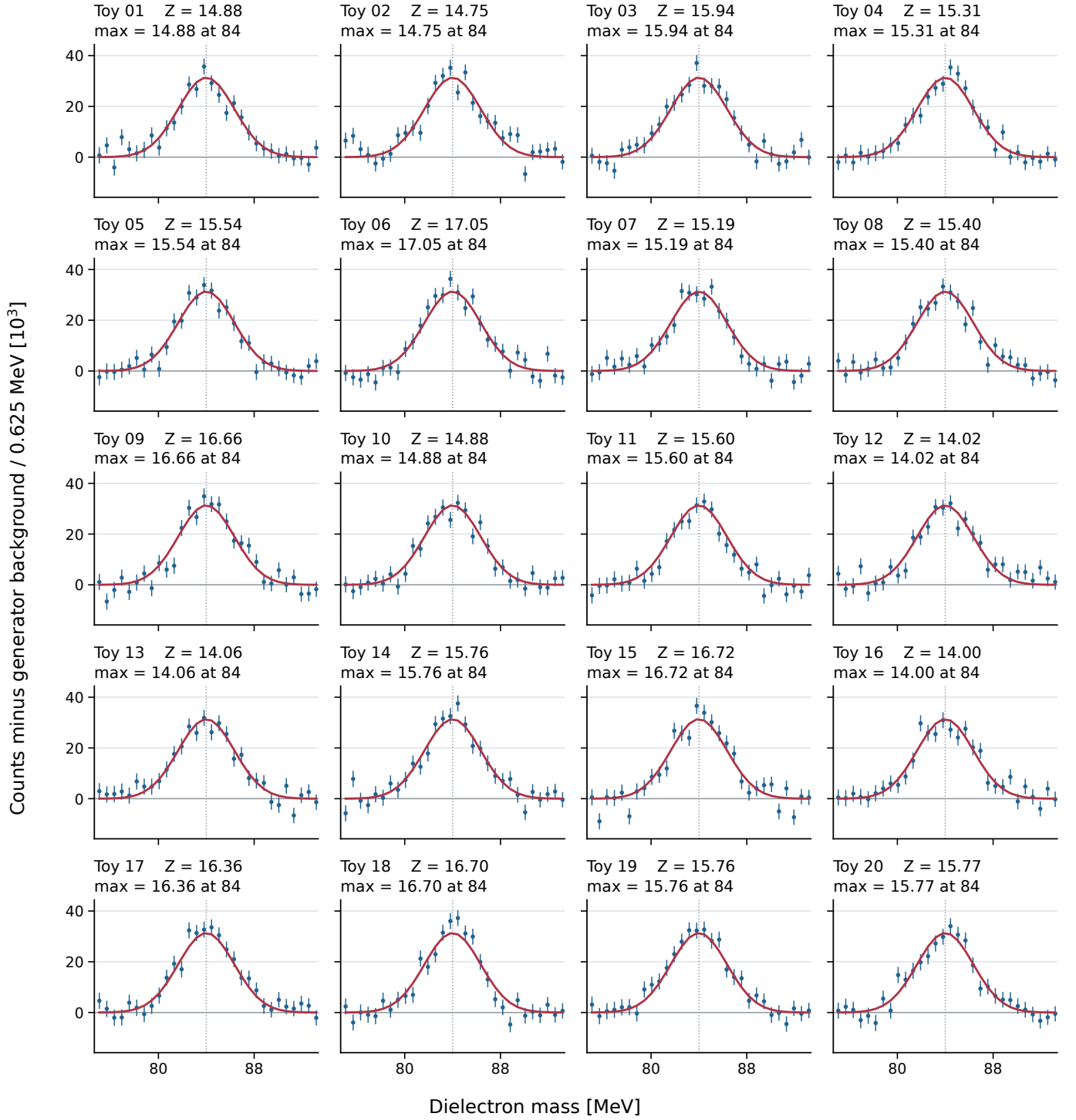
Shaded bands mark the $\pm 2.25\sigma_m$ extraction window. Observed error bars are $\sqrt{n}$; the dotted Asimov excess subtracts the profiled background in that window. The lower-right curves use fresh GP conditioning at every tested mass.

Yield-scaled amplitude a_{yield}	270,236	Matched amplitude a_{match}	295,028
Yield-scaled Asimov Z	14.046	Amplitude ratio R_a	1.092
Background-only Asimov Z	0.088	Matched Asimov $Z(m_0)$	15.326
Signal inside fit window	288,079	Background inside fit window	169,032,890
Fixed-mass median Z	15.57	Empirical 16–84% interval	[14.75, 16.65]
Largest fixed-mass Z	17.05	Largest nearby-maximum Z	17.05
Fixed-mass $Z \geq 3$	20/20	Fixed-mass $Z \geq 5$	20/20

The nearby scan covers 80–88 MeV in 1 MeV steps (9 hypotheses). Its median maximum is 15.57; 0/20 toy maxima lie at a grid endpoint. The matched Asimov maximum is 15.33 at 84 MeV.

Additional 1% peak: 84 MeV: 20 realizations

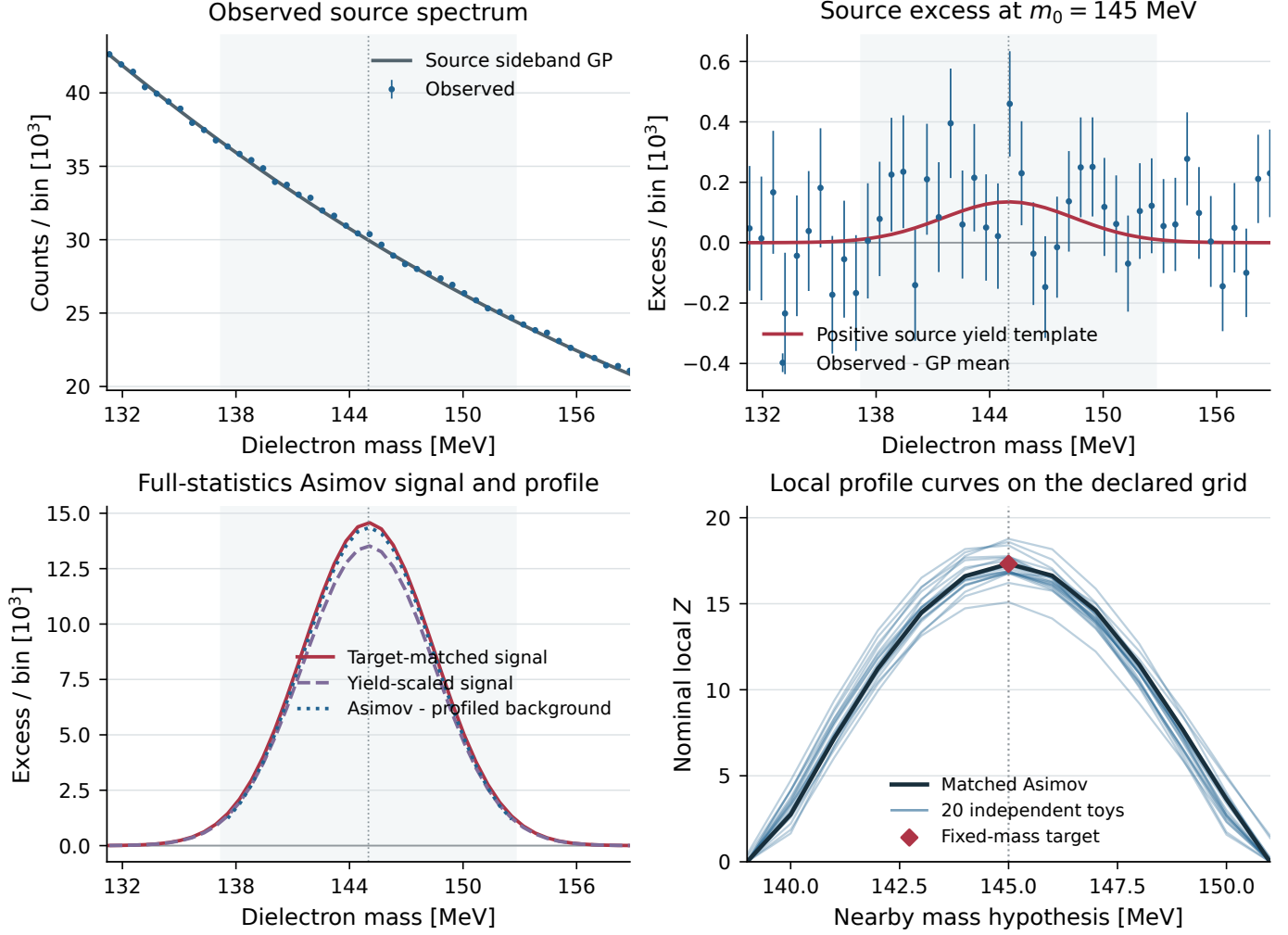
$1\% \times 100$; $m_0 = 84$ MeV; $Z_{\text{target}} = 15.33$. Red curves show the injected signal; points show counts minus generator background.



All 20 independent Poisson draws appear in saved seed order (display toy 01 is stored ID 0). Every panel uses identical axes within this scenario and $\sqrt{n}$ count errors. “ Z ” is evaluated at the injection mass; “max” scans the declared nearby hypotheses, with its mass in MeV. An asterisk marks a grid-endpoint maximum.

Additional 1% peak: 145 MeV

Injection at $m_0 = 145$ MeV, $\sigma_m = 3.454$ MeV; source $Z_f = 1.732$, requested $Z_{\text{target}} = 17.319$.



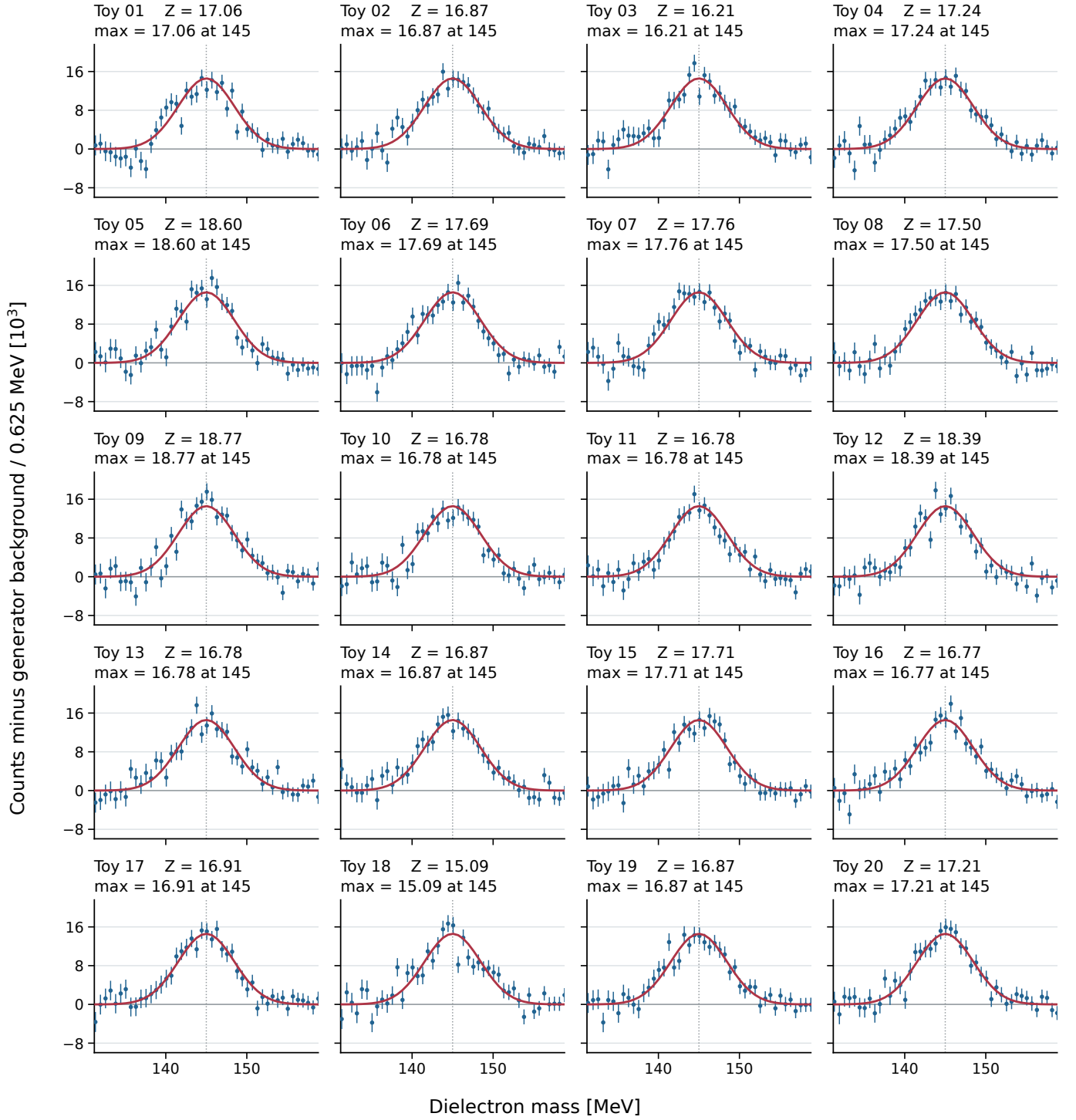
Shaded bands mark the $\pm 2.25\sigma_m$ extraction window. Observed error bars are $\sqrt{n}$; the dotted Asimov excess subtracts the profiled background in that window. The lower-right curves use fresh GP conditioning at every tested mass.

Yield-scaled amplitude a_{yield}	187,640	Matched amplitude a_{match}	202,218
Yield-scaled Asimov Z	16.074	Amplitude ratio R_a	1.078
Background-only Asimov Z	0.036	Matched Asimov $Z(m_0)$	17.319
Signal inside fit window	197,416	Background inside fit window	75,281,386
Fixed-mass median Z	16.98	Empirical 16–84% interval	[16.78, 17.76]
Largest fixed-mass Z	18.77	Largest nearby-maximum Z	18.77
Fixed-mass $Z \geq 3$	20/20	Fixed-mass $Z \geq 5$	20/20

The nearby scan covers 139–151 MeV in 1 MeV steps (13 hypotheses). Its median maximum is 16.98; 0/20 toy maxima lie at a grid endpoint. The matched Asimov maximum is 17.32 at 145 MeV.

Additional 1% peak: 145 MeV: 20 realizations

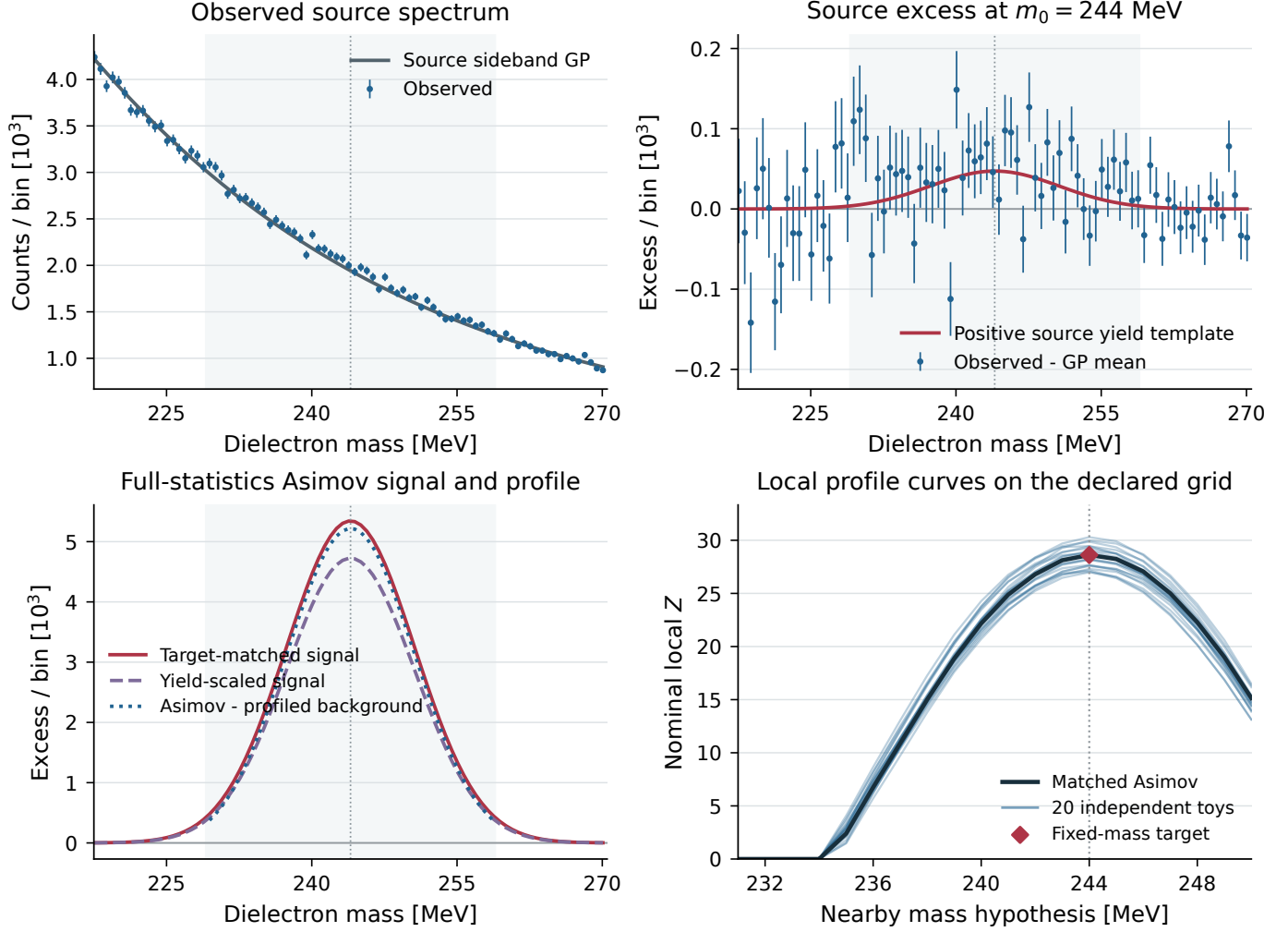
$1\% \times 100$; $m_0 = 145$ MeV; $Z_{\text{target}} = 17.32$. Red curves show the injected signal; points show counts minus generator background.



All 20 independent Poisson draws appear in saved seed order (display toy 01 is stored ID 0). Every panel uses identical axes within this scenario and $\sqrt{n}$ count errors. “Z” is evaluated at the injection mass; “max” scans the declared nearby hypotheses, with its mass in MeV. An asterisk marks a grid-endpoint maximum.

Additional 1% peak: 244 MeV

Injection at $m_0 = 244$ MeV, $\sigma_m = 6.625$ MeV; source $Z_f = 2.861$, requested $Z_{\text{target}} = 28.614$.



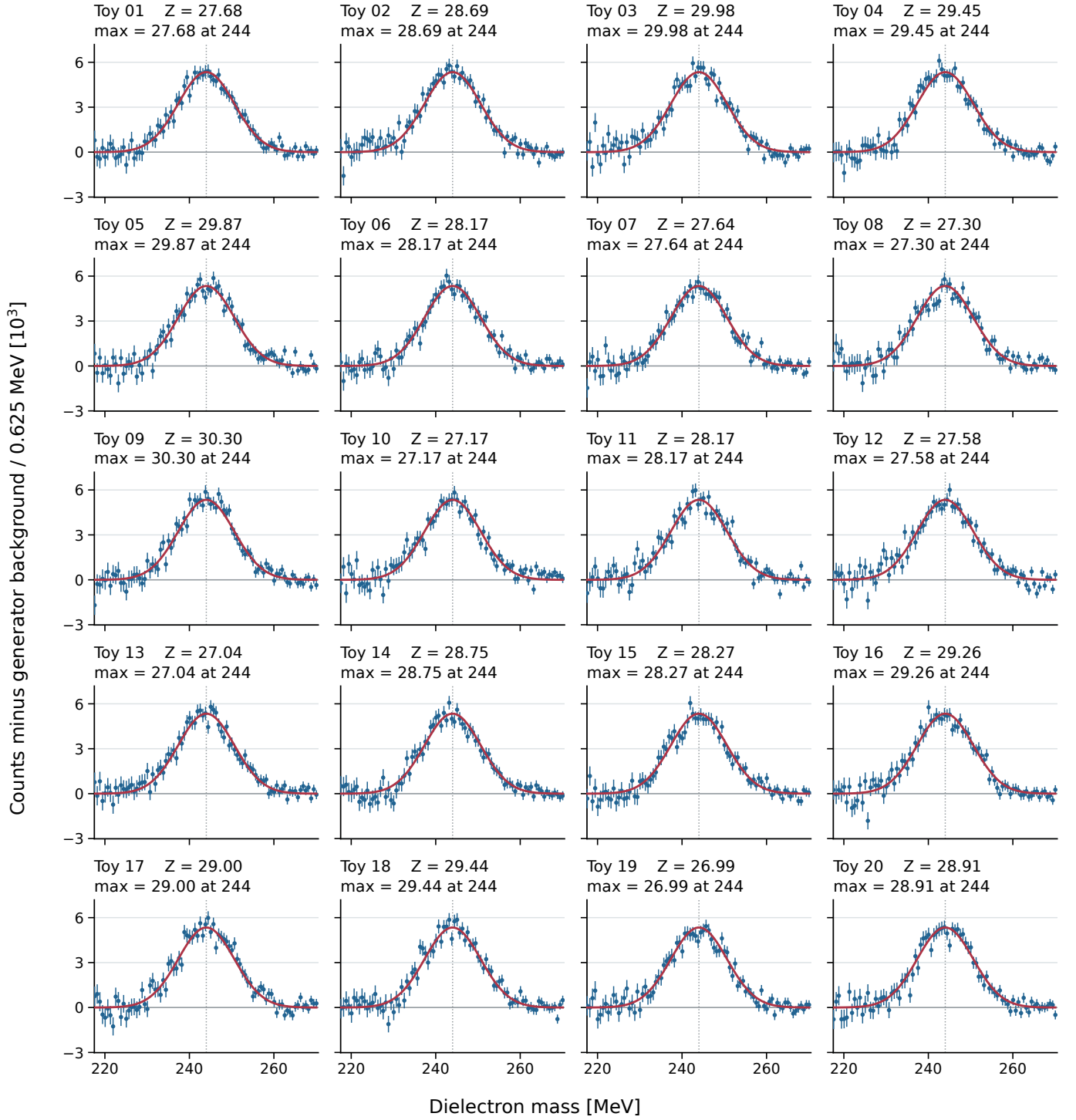
Shaded bands mark the $\pm 2.25\sigma_m$ extraction window. Observed error bars are $\sqrt{n}$; the dotted Asimov excess subtracts the profiled background in that window. The lower-right curves use fresh GP conditioning at every tested mass.

Yield-scaled amplitude a_{yield}	125,607	Matched amplitude a_{match}	142,064
Yield-scaled Asimov Z	25.313	Amplitude ratio R_a	1.131
Background-only Asimov Z	0.000	Matched Asimov $Z(m_0)$	28.614
Signal inside fit window	138,711	Background inside fit window	9,598,751
Fixed-mass median Z	28.48	Empirical 16–84% interval	[27.31, 29.45]
Largest fixed-mass Z	30.30	Largest nearby-maximum Z	30.30
Fixed-mass $Z \geq 3$	20/20	Fixed-mass $Z \geq 5$	20/20

The nearby scan covers 231–250 MeV in 1 MeV steps (20 hypotheses). Its median maximum is 28.48; 0/20 toy maxima lie at a grid endpoint. The matched Asimov maximum is 28.61 at 244 MeV.

Additional 1% peak: 244 MeV: 20 realizations

$1\% \times 100$; $m_0 = 244$ MeV; $Z_{\text{target}} = 28.61$. Red curves show the injected signal; points show counts minus generator background.



All 20 independent Poisson draws appear in saved seed order (display toy 01 is stored ID 0). Every panel uses identical axes within this scenario and $\sqrt{n}$ count errors. “Z” is evaluated at the injection mass; “max” scans the declared nearby hypotheses, with its mass in MeV. An asterisk marks a grid-endpoint maximum.

Reproducibility and numerical record

The report is generated from saved arrays and tables, without performing new fits. The package preserves both source histories and the chosen model state. The full SHA-256 ledgers identify individual inputs and outputs; inherited manifest entries also describe parent-release files and should not be read as a list of files copied into this study.

Pinned computational inputs

Input	SHA-256
Exported lane spectra	aa993417cab1f3a309bf92913743c391d4ba1c5a4bb038ffd047d49ad615a8ea
Pinned 10% spectrum and states	fd4867a7e2df69d88d62cdaaed133096d015371e2fe5bd700596f411c8437294
Historical reviewed 1% states	a3b3c4feac9a0ce8be07329514b3696bd7e8505a1cd874bf660f6afeebe7475f
Historical last-use protocol	3a68c17913aa16567605399d91c14fb2ffe03aaaa6ccfa98084044514cbbd219

The original 1% ROOT histogram is `preselection/h_invM_8000` in `final_1pct_invM.root`; its recorded SHA-256 is `eeeb5b8f40820c8f3c5d9673d2cddc155f2e2282474e20eb4f8930fd0a561ada`. The last-use protocol is dated 14 August 2026. Historical reviewed states preserve LML-selected repeat provenance; `inputs/historical_one/PROVENANCE.md` and `copy_manifest.json` give the source paths and exact byte identities. `inputs/lane_manifest.json` records the remaining exposure and selection limitations.

Saved products and deterministic rebuilding

<code>derived/source_scan.csv</code>	Both observed 201-point scans.
<code>derived/selected_peaks.csv</code>	Source selection, target and scenario identity.
<code>derived/catalogue.csv</code>	Amplitudes, Asimov comparisons and toy summaries.
<code>toys/<scenario>.npz</code>	All counts, truth, source spectrum, masks and fitted arrays.
<code>derived/<scenario>_curves.csv</code>	Toy and Asimov scores at every nearby hypothesis.
<code>derived/<scenario>_toys.csv</code>	Fixed-mass and nearby-max results for each toy ID.
<code>derived/report_display_ledger.json</code>	Display ranges and scan-boundary identifications.

Each toy uses NumPy `SeedSequence` with $[56020210912, K, t]$, where $t = 0, \dots, 19$, $K = \text{round}(1000m_0)$ for the 1% lane and $K = 10^6 + \text{round}(1000m_0)$ for the 10% lane, with m_0 in MeV. One worker and one linear-algebra thread are used. Saved toys are not replaced on the basis of their fitted score. From the package root, run `pythonscripts/run_study.pyprepare`, then `pythonscripts/run_study.pytoys`, then `pythonscripts/validate_study.py`, then `pythonscripts/make_report.py`. Compile `source/main.tex` from its own directory with Tectonic or pdfLaTeX. The matching ROOT input is included. Existing scenario caches are reused only after their dependency fingerprints are verified.

Numerical status

The log-GP uses the positive eigenspace of the RBF Gram matrix with a relative eigenvalue cutoff of 10^{-14} and SVD ridge conditioning, avoiding subtraction of nearly equal posterior-covariance matrices. Target matching requires a background-only score below the requested target and an achieved absolute error below 10^{-5} ; a failed gate stops the scenario. The runner completed 15 scenarios and 300 toys. The largest absolute target-matching error is $7.64e - 11$; the largest fit score residual is $2.84e - 07$. All recorded fitted background and total means are positive. These checks establish numerical consistency of the saved conditional fits; they do not calibrate significance tails. An independent audit passed 342/342 checks, including replay of saved seeds, complete scenario and scan grids, and source hashes. Eight GP precision comparisons change Z by at most 1.62×10^{-4} . Source-score differences from archived studies are recorded in `derived/source_archived_comparison.csv`; they are method comparisons, not exposure effects.

References

- [1] G. Cowan, K. Cranmer, E. Gross and O. Vitells, *Asymptotic formulae for likelihood-based tests of new physics*, Eur. Phys. J. C **71**, 1554 (2011); erratum **73**, 2501 (2013). arXiv:1007.1727. Asimov data and discovery statistics: Sections 2.3, 3.5, 4.1 and 5.1.1.